



BANGLADESH UNIVERSITY OF ENGINEERING & TECHNOLOGY

CHE 400

Project/Thesis I

A Final report on

Ca-Fe Oxide on Carbon Catalyst for Bio-diesel Production

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Submission Date:

8 June 2026

Acknowledgment

The authors express their sincere gratitude to Dr. Kazi Bayzid Kabir, Professor, Department of Chemical Engineering, Bangladesh University of Engineering and Technology (BUET), for proposing this valuable research topic and for his continuous guidance, encouragement, and insightful suggestions throughout the course of this study. His expertise, supervision, and constructive feedback played a crucial role in the successful completion of this research work.

The authors would also like to extend their heartfelt appreciation to the faculty members and staff of the Department of Chemical Engineering, BUET, for providing the necessary academic support, laboratory facilities, and technical assistance required for conducting the research. Special thanks are extended to the laboratory personnel and technicians who assisted in various experimental activities and ensured the smooth operation of laboratory equipment during the study.

The authors are grateful to the personnel involved in sample characterization and analytical testing for their cooperation and valuable support in conducting XRD and other necessary analyses. Their assistance greatly contributed to the successful evaluation and interpretation of the research findings.

The authors would also like to acknowledge all individuals who directly or indirectly contributed to this work through their advice, cooperation, and encouragement throughout the research period.

Abstract

The increasing demand for sustainable and environmentally friendly energy sources has stimulated significant interest in biodiesel as a renewable alternative to conventional diesel fuel. The issues associated with the recovering of catalysts, purification of the products and wastewater generation are, however, challenges in the use of homogeneous catalysts in biodiesel production. This study aims to investigate the synthesis and use of a $\text{CaO-Fe}_3\text{O}_4$ catalyst on activated carbon to produce biodiesel from soybean oil by transesterification process.

Fe_3O_4 nanoparticles were synthesized via a co-precipitation method using ferrous sulfate heptahydrate and ferric nitrate nonahydrate as precursor materials. The synthesized Fe_3O_4 was subsequently combined with calcium nitrate and activated carbon through a wet impregnation technique followed by calcination at 500°C under a nitrogen atmosphere to produce $\text{CaO-Fe}_3\text{O}_4/\text{activated carbon}$ catalysts. Three different compositions of catalysts were prepared and used with the C/O ratios of 90:10, 80:20 and 70:30. The soybean oil was used as a raw material for biodiesel production, while methanol was used as a solvent to react with the oil. The synthesized catalyst was analyzed by X-ray diffraction (XRD) and produced biodiesel was analyzed by density, cloud point, pour point, CHNS elemental analysis and higher heating value (HHV) determination.

The XRD spectra has also confirmed the presence of crystalline Fe_3O_4 phases with characteristic diffraction peaks at 35.95° , 43.15° and 57.30° (2θ). In addition, peaks for CaCO_3 were also detected, suggesting that CaO was partially carbonated from atmospheric exposure. The average biodiesel production from the three catalyst compositions (90:10, 80:20 and 70:30) were found to be 60.8%, 78.2% and 90.4%, respectively, which indicates that the transesterification performance improved with increase in oxide content. The densities of the produced biodiesel were in the range $904.74\text{--}907.89\text{ kg/m}^3$, cloud point was in range -4°C to 1°C , pour point was in range -10°C to -6.5°C , carbon content obtained by the CHNS analysis was 73.16–74.30% and hydrogen content obtained by the CHNS analysis was 10.96–11.40%, and calculated HHV values were 36.98–37.91 MJ/kg. The 70:30 ratio of carbon to oxides gave the best yield of biodiesel while 80:20 ratio gave the most favorable cold-flow properties among the catalyst formulations investigated. These results indicate that $\text{CaO-Fe}_3\text{O}_4$ supported on an activated carbon is a promising heterogeneous catalyst for biodiesel production and that the performance of the catalyst and the quality of the biodiesel greatly depends on the composition of the catalyst.

Table of Contents

Contents

Acknowledgment	1
Abstract	2
List of Tables	6
1 Introduction.....	1
1.1 Background of the study.....	1
1.1.1 Global Energy Demand and Environmental Concerns	1
1.1.2 Biodiesel as a Renewable Alternative Fuel.....	1
1.1.3 Biodiesel Production through Transesterification.....	2
1.2 Problem Statement	3
1.3 Research Objectives	3
1.3.1 General Objective	3
1.3.2 Specific Objectives	3
1.4 Scope of the study	4
1.5 Significance of the study	5
1.6 Overview of Research Methodology.....	5
2 Literature Review.....	6
2.1 Introduction	6
2.2 Biodiesel.....	6
2.2.1 Definition and Properties	7
2.2.2 Feedstocks for Biodiesel Production.....	7
2.3 Transesterification Process and Catalysts	9
2.4 Heterogeneous Catalysts in Biodiesel Production	11
2.5 Metal Oxide Catalysts for Biodiesel Production.....	12
2.6 Carbon-Supported Catalysts	14
2.7 Critical Analysis of Previous Studies and Research Gap	15
3 Materials And Methodology	17
3.1 Introduction	17
3.2 Research Design.....	17
3.2.1 Experimental Framework.....	17
3.2.2 Methodological Flowchart	18
3.3 Materials.....	19
3.3.1 Chemical & Reagents.....	19
3.4 Catalyst Synthesis.....	19
3.4.1 Synthesis of Fe ₃ O ₄ nanoparticles	19

3.4.2	Preparation of CaO-Fe ₃ O ₄ Precursor Mixture	20
3.4.3	Catalyst Impregnation Procedure	21
3.4.4	Catalyst Calcination	22
3.5	Catalyst Composition Variation	23
3.6	Biodiesel Production Experiment.....	23
3.6.1	Experimental Setup	23
3.6.2	Transesterification Procedure	23
3.6.3	Catalyst Separation and Product Recovery	24
3.7	Catalyst Characterization	25
3.7.1	X-Ray Diffraction (XRD) Analysis	25
3.8	Biodiesel Characterization	26
3.8.1	Biodiesel Yield Calculation	26
3.8.2	Density Measurement (ASTM D941).....	26
3.8.3	Cloud Point (ASTM D2500).....	27
3.8.4	Pour Point (ASTM D97).....	27
3.8.5	CHNS Elemental Analysis.....	28
3.8.6	Higher Heating Value (HHV) Determination	28
3.9	Data Analysis	30
4	Result and Discussion	31
4.1	Introduction	31
4.2	Catalyst Characterization Results.....	31
4.2.1	XRD Analysis	31
4.2.2	Catalyst Yield Analysis.....	35
4.3	Biodiesel Production Results.....	36
4.3.1	Biodiesel yield analysis based on the effect of activated carbon-oxides ratio.....	37
4.4	Biodiesel Quality Assessment.....	38
4.4.1	Fuel Properties Comparison with Standards	38
4.4.2	Higher heating value analysis	40
4.5	Comparison with Previous Studies.....	40
4.5.1	Comparison of Yield.....	41
4.5.2	Correlation with Conceptual Framework.....	42
4.6	Limitations of Findings	43
4.6.1	Experimental Limitations.....	43
4.6.2	Material Limitations.....	44
4.6.3	Analytical Limitations.....	44
4.7	Mitigation Strategies and Improvements	45
4.7.1	Improvements in experimental design	45

4.7.2	Improvements in Catalyst Preparation	45
4.7.3	Improvements in Analytical Methods	46
5	Impact and Implications	47
5.1	Introduction	47
5.2	Environmental Impact	47
5.3	Economic and Industrial Implications.....	48
5.4	Health and Safety Considerations	49
5.5	Sustainability Aspects	50
6	Conclusion and Recommendations	51
6.1	Introduction	51
6.2	Summary of Key findings	51
6.3	Conclusions	52
6.4	Recommendations	53
6.5	Scope for Future Work	53
	References	55
	Appendix.....	57
	Appendix A : Mapping of the research problem with the attributes of complex engineering problems.....	57
	Appendix B : Research plan (preliminary and revised)Preliminary Work Plan.....	59
	Appendix C : Risk Assessment.....	61
	Appendix D : Environmental Mitigation and Monitoring Plan	65

List Of Figures

Figure 1: Benefits of biodiesel over petroleum diesel	2
Figure 2: Types of Biodiesel Feedstock.....	8
Figure 3: (a) Overall transesterification reaction, and (b) transesterification reaction (sequential reaction)	9
Figure 4: Flow diagram of the overall experimental methodology.	18
Figure 5: Schematic diagram of the Fe ₃ O ₄ nanoparticle synthesis setup.....	20
Figure 6: Schematic diagram of the CaO-Fe ₃ O ₄ precursor mixture through wet impregnation.	21
Figure 7: Schematic diagram of the calcination setup.	22
Figure 8: Schematic diagram of the transesterification reaction setup.	24
Figure 9: Schematic diagram of the filtration setup for recovery of catalyst & product.	25
Figure 10: Pre-Transesterification Reaction peak identification	32
Figure 11: Post-Transesterification Reaction peak identification.....	33

List of Tables

Table 1: Different type of chemicals, formula and purpose used in the experiment	19
Table 2: Catalyst Yield Analysis	35
Table 3: Biodiesel Yield analysis.....	37
Table 4: Knowledge profile of the research.....	57
Table 5: Complex engineering problem solving.....	58
Table 6: Table For the risk assessment of the project.....	62
Table 7: Table for the environmental mitigation and monitoring plan.....	65

1 Introduction

1.1 Background of the study

1.1.1 Global Energy Demand and Environmental Concerns

In the present world are predominantly dependent on energy for major economic sectors, primarily transportation and shipping services, which are expected to rise by over 50% by the year 2030. Rapid globalization and an exponentially rising world population have caused an escalation in energy demand at a fast pace, thus resulting in excessive energy consumption. Currently, fossil fuels provide 80% of the energy used in the world of which, around 96% of the shipping and transportation division relies on fossil fuels comprising petroleum, coal, crude oil and natural gas. That is how the extravagant expenditure on fossil-derived fuels has been taking place causing a major decline in fossil fuels. The depletion of these resources leads to instability in fuel prices harming the economy of the nation and bringing severe environmental concern, as their combustion discharges toxic gases like hydrocarbons, CO, particulate matter and oxides of N, along with CO₂. Over the past several years, scientists have been exploring the development of biofuels that can function as befitting supplements to conventional petro-diesel. Biofuels are sourced from the biomass of living beings such as plants, animals and microorganisms that act as energy-rich renewable resources. Among the diverse generations of biofuels, biodiesel has gained the utmost popularity. [1]

1.1.2 Biodiesel as a Renewable Alternative Fuel

Notably, biodiesel stands out as a significant player in the realm of renewable fuels, boasting numerous advantages over Petro diesel (also known as diesel or petroleum diesel), as depicted in Figure 1. The integration of heterogeneous catalysts holds promise in curbing the substantial production costs associated with biodiesel, thereby enhancing its competitiveness with Petro diesel fuels. The data presented in Figure 1, unmistakably highlights biodiesel's superior qualities, including its enhanced lubricity, higher cetane numbers, reduced GHG emissions, lower toxicity, improved biodegradability, and its ability to enhance air quality while fulfilling energy demands. Given the limitations of current internal combustion (IC) motors to directly utilize 100% biodiesel, a practical approach involves blending biodiesel with conventional diesel fuel. As depicted in Figure 1, biodiesel exhibits superior safety in terms of transportation, storage, and environmental impact compared to petroleum diesel. Notably, diesel and biodiesel have flash values of 60°C and 150°C, respectively, further solidifying biodiesel's potential as a viable diesel substitute. With an oxygen content of 11%,

biodiesel's structural composition aids in more complete combustion, resulting in decreased emissions of carbon monoxide (CO) and unburned hydrocarbons when compared to traditional diesel fuels. This inherent characteristic underscores the safety advantage of using biodiesel over diesel. [2]



Figure 1: Benefits of biodiesel over petroleum diesel

1.1.3 Biodiesel Production through Transesterification

Biodiesel is most commonly produced through the transesterification process, in which triglycerides present in vegetable oils, animal fats, or waste oils react with a short-chain alcohol, typically methanol, in the presence of a catalyst. This reaction converts the triglycerides into fatty acid methyl esters (FAME), known as biodiesel, while glycerol is generated as a by-product. Transesterification is widely preferred due to its relatively high conversion efficiency, mild operating conditions, and ability to produce biodiesel with properties similar to conventional diesel fuel.[3]

The performance of the transesterification process is influenced by several factors, including the alcohol-to-oil ratio, catalyst type and concentration, reaction temperature, and reaction time. In recent years, heterogeneous catalysts have attracted considerable attention because they offer easier

separation, reusability, and reduced environmental impact compared with conventional homogeneous catalysts. [4]

1.2 Problem Statement

Biodiesel has emerged as a promising renewable alternative to conventional diesel fuel due to its biodegradability, lower emissions, and sustainable production from renewable feedstocks. The transesterification process is widely used for biodiesel production; however, its efficiency largely depends on the catalyst employed. Conventional homogeneous catalysts, such as sodium hydroxide and potassium hydroxide, provide high reaction rates but suffer from several drawbacks, including difficult catalyst recovery, wastewater generation, soap formation, and poor reusability. These limitations increase production costs and hinder the sustainability of biodiesel manufacturing.[5]

Heterogeneous catalysts have attracted significant attention because they can be easily separated and reused. Among them, calcium oxide (CaO) exhibits high catalytic activity and low cost, making it a promising catalyst for biodiesel production.[6] However, CaO-based catalysts often experience leaching, deactivation, and reduced stability during repeated use, which negatively affect biodiesel yield and catalyst lifespan. Previous studies have suggested that modifying CaO with supporting it on different materials can improve catalyst stability, surface area and performance. In this thesis, an investigation is done to evaluate the effectiveness of carbon-supported Ca-Fe oxide catalysts for biodiesel production.

1.3 Research Objectives

1.3.1 General Objective

To synthesize and evaluate a carbon-supported Ca-Fe oxide heterogeneous catalyst for biodiesel production.

1.3.2 Specific Objectives

1. To prepare a Ca-Fe oxide/Activated carbon catalyst.
2. To characterize the synthesized catalyst.
3. To evaluate its performance in biodiesel production.
4. To investigate the effects of activated carbon and oxides ratio on biodiesel yield.

1.4 Scope of the study

This study focuses on the synthesis, characterization, and performance evaluation of a carbon-supported Ca-Fe oxide heterogeneous catalyst for biodiesel production via the transesterification process. The catalyst is prepared using selected laboratory-scale methods and evaluated under controlled experimental conditions to determine its suitability for biodiesel synthesis.

The scope of the study includes the preparation of the Ca-Fe oxide/carbon catalyst, its physicochemical characterization using techniques such as XRD, BET and its application in transesterification of a selected oil feedstock using methanol. The influence of a key reaction parameter activated carbon and oxides ratio is investigated to determine their effect on biodiesel yield.

This study is limited to laboratory-scale experiments and does not include pilot-scale or industrial-scale production. Economic feasibility analysis, life cycle assessment, and long-term industrial performance evaluation are also beyond the scope of this research.

1.5 Significance of the study

This study is significant as it contributes to the development of efficient and sustainable heterogeneous catalysts for biodiesel production. The use of a carbon-supported Ca-Fe oxide catalyst is expected to enhance catalytic performance by improving surface area, active site dispersion, and overall stability, thereby addressing common limitations associated with conventional catalysts such as leaching and poor reusability. From an environmental perspective, the research supports the advancement of renewable energy technologies by promoting biodiesel as a cleaner alternative to fossil fuels. Improved catalyst efficiency can lead to higher biodiesel yields under milder conditions, thereby reducing energy consumption and associated emissions during production. From an industrial and economic point of view, the development of a low-cost, reusable catalyst has the potential to reduce biodiesel production costs and improve process sustainability. The findings of this study may also provide useful insights for scaling up catalyst design and optimizing biodiesel production systems in future applications.

1.6 Overview of Research Methodology

This study adopts an experimental laboratory-scale approach to develop and evaluate a carbon-supported Ca-Fe oxide heterogeneous catalyst for biodiesel production. The research begins with the synthesis of Fe_3O_4 nanoparticles through a co-precipitation method using iron salts and ammonium hydroxide under controlled conditions. Subsequently, $\text{CaO-Fe}_3\text{O}_4$ is prepared and supported on activated carbon via wet impregnation followed by drying and calcination under a nitrogen atmosphere to obtain the final catalyst. The prepared catalyst is characterized to determine its structural, morphological, and physicochemical properties using standard analytical techniques such as X-ray diffraction (XRD), surface area analysis (BET), where applicable. These analyses are used to confirm the formation of active phases and evaluate catalyst properties relevant to performance. The catalytic activity is evaluated through transesterification of soybean oil using methanol under atmospheric pressure in a batch reactor. After reaction completion, the catalyst is separated by filtration, and the product mixture is subjected to centrifugation to separate biodiesel from glycerol. Finally, biodiesel yield is determined, and the catalyst is assessed for its effect before and after transesterification reaction. The overall methodology enables evaluation of the relationship between catalyst properties and biodiesel production performance under optimized laboratory conditions.

2 Literature Review

2.1 Introduction

The increasing demand for sustainable energy and the environmental impacts of fossil fuel consumption have driven extensive research into renewable fuels such as biodiesel. Biodiesel is considered a promising alternative due to its biodegradability, lower emissions, and compatibility with existing diesel engines. Its production mainly relies on catalytic transesterification of triglyceride-based feedstocks. The efficiency of this process strongly depends on the type of catalyst used, which has led to significant interest in developing advanced heterogeneous catalysts with high activity, stability, and reusability.

2.2 Biodiesel

Biodiesel has emerged as one of the most promising renewable alternatives to conventional petroleum-derived diesel fuel due to increasing concerns regarding energy security, environmental pollution, and climate change. It is produced from renewable biological resources such as vegetable oils, animal fats, and waste cooking oils, making it a sustainable substitute for fossil fuels. Biodiesel is attractive because it can be utilized in existing diesel engines with little or no modification while significantly reducing emissions of sulfur compounds, particulate matter, carbon monoxide, and unburned hydrocarbons.

The growing demand for renewable transportation fuels has accelerated global biodiesel research and commercialization. Compared with fossil diesel, biodiesel offers several environmental advantages, including biodegradability, lower toxicity, and reduced greenhouse gas emissions throughout its life cycle. Additionally, biodiesel contributes to energy diversification by reducing dependence on imported petroleum resources. These benefits have encouraged governments worldwide to implement renewable fuel standards and biodiesel blending mandates to promote its adoption.

Despite these advantages, biodiesel production faces several challenges, including high feedstock costs, catalyst efficiency, fuel stability, and large-scale economic feasibility. Feedstock costs alone can account for a substantial portion of total production expenses, making process optimization and catalyst development critical areas of research. Consequently, significant efforts have been devoted to developing efficient heterogeneous catalysts capable of improving biodiesel yield while reducing production costs and environmental impact.

2.2.1 Definition and Properties

Biodiesel is defined as a fuel composed of mono-alkyl esters of long-chain fatty acids derived from renewable lipid sources such as vegetable oils, animal fats, and waste oils. It is commonly produced through the transesterification of triglycerides with short-chain alcohols, typically methanol, resulting in fatty acid methyl esters (FAME) and glycerol as a by-product. This definition has been widely accepted by international fuel standards such as ASTM D6751 and EN 14214.

One of the most significant characteristics of biodiesel is its renewable nature. Unlike fossil diesel, which originates from finite petroleum reserves, biodiesel is produced from continuously replenishable biological resources. Biodiesel also possesses excellent lubricity, which helps reduce engine wear and improve fuel system performance. Furthermore, it has a higher flash point than petroleum diesel, making it safer to handle and transport. [7]

The physicochemical properties of biodiesel depend strongly on the fatty acid composition of the feedstock used. Important fuel properties include viscosity, density, cetane number, oxidative stability, cold-flow behavior, and calorific value. Generally, biodiesel exhibits higher viscosity and density than petroleum diesel, which can influence fuel atomization and combustion performance. However, its oxygen content promotes more complete combustion, resulting in lower emissions of carbon monoxide, hydrocarbons, and particulate matter.

The fatty acid structure significantly affects biodiesel quality. Biodiesel produced from feedstocks rich in saturated fatty acids generally exhibits higher cetane numbers and oxidation stability but poorer cold-flow properties. Conversely, biodiesel derived from unsaturated fatty acids typically demonstrates improved cold-flow characteristics but lower oxidation resistance. Therefore, feedstock selection plays a crucial role in determining final fuel quality. [8]

2.2.2 Feedstocks for Biodiesel Production

Feedstock selection is one of the most critical factors influencing biodiesel production economics and fuel quality. Various feedstocks have been investigated for biodiesel synthesis, including edible vegetable oils, non-edible oils, animal fats, algal oils, and waste cooking oils. The choice of feedstock directly affects biodiesel yield, production cost, fuel properties, and overall sustainability.

Edible vegetable oils such as soybean oil, palm oil, rapeseed oil, and sunflower oil are among the most commonly used feedstocks because of their high oil content and established agricultural supply

chains. Soybean oil is particularly popular in North America, whereas palm oil dominates biodiesel production in Southeast Asia. These feedstocks generally produce high-quality biodiesel with favorable fuel properties and conversion efficiencies.

However, the use of edible oils has generated concerns regarding food security and land utilization, commonly referred to as the "food-versus-fuel" debate. As a result, increasing attention has been directed toward non-edible oils such as jatropha, castor, and karanja oils. These feedstocks can be cultivated on marginal land and do not directly compete with food production.

Waste cooking oils and animal fats represent another important category of biodiesel feedstocks. Their utilization not only reduces production costs but also addresses waste management challenges. However, these feedstocks often contain higher levels of free fatty acids and impurities, requiring additional pretreatment steps before transesterification. [9]

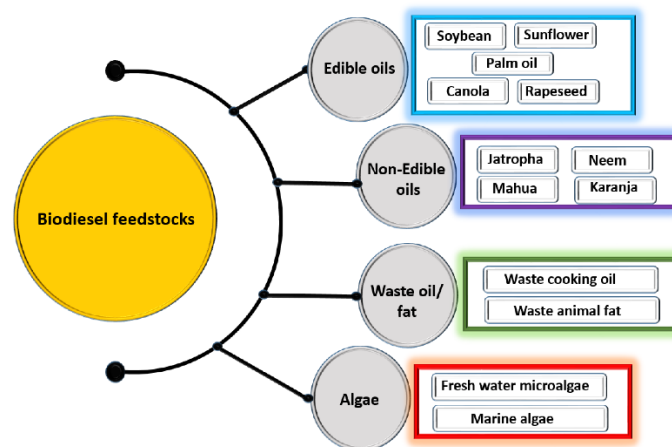


Figure 2: Types of Biodiesel Feedstock[4]

2.3 Transesterification Process and Catalysts

Transesterification is the most widely adopted method for biodiesel production due to its high conversion efficiency, relatively mild operating conditions, and economic feasibility. The process involves the reaction of triglycerides present in vegetable oils, animal fats, or waste oils with a short-chain alcohol, typically methanol, in the presence of a catalyst. During the reaction, triglycerides are converted into fatty acid methyl esters (FAME), commonly known as biodiesel, while glycerol is produced as a valuable by-product. Compared with other biodiesel production methods such as pyrolysis and microemulsion, transesterification offers superior fuel quality, lower viscosity, and higher conversion rates. [10]

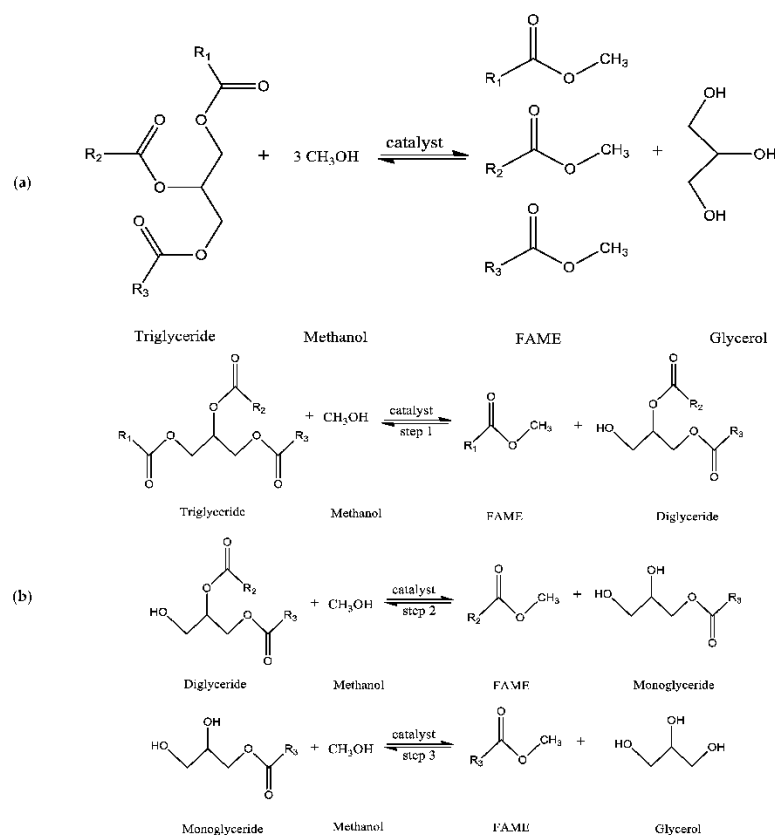


Figure 3: (a) Overall transesterification reaction, and (b) transesterification reaction (sequential reaction).[4]

The transesterification reaction proceeds through three consecutive reversible steps. Initially, triglycerides react with alcohol to form diglycerides and one molecule of fatty acid ester. The diglycerides subsequently react to form monoglycerides and another ester molecule. Finally, monoglycerides are converted into glycerol and the third ester molecule. Since the reaction is reversible, excess alcohol is generally used to shift the equilibrium toward biodiesel formation and

maximize conversion. The methanol-to-oil molar ratio, catalyst concentration, reaction temperature, and reaction time are among the most important parameters influencing biodiesel yield. [11]

Catalysts play a crucial role in enhancing the reaction rate by lowering the activation energy and improving contact between the immiscible oil and alcohol phases. Biodiesel catalysts are generally classified into homogeneous, heterogeneous, and enzymatic catalysts. Homogeneous catalysts, including sodium hydroxide (NaOH), potassium hydroxide (KOH), and sulfuric acid (H₂SO₄), have been extensively used in commercial biodiesel production because of their high catalytic activity and rapid reaction rates. However, these catalysts suffer from several disadvantages, including catalyst recovery difficulties, soap formation, wastewater generation, and sensitivity to free fatty acids and moisture content. These drawbacks increase downstream purification requirements and overall production costs. [11]

To overcome these limitations, heterogeneous catalysts have gained significant attention in recent years. Unlike homogeneous catalysts, heterogeneous catalysts exist in a different phase from the reactants, usually as solid materials suspended in liquid reactants. Their major advantages include ease of separation, catalyst recyclability, reduced wastewater generation, and improved environmental sustainability. Various heterogeneous catalysts have been investigated for biodiesel production, including metal oxides, zeolites, hydrotalcites, ion-exchange resins, and carbon-supported catalysts. Among these materials, metal oxide catalysts such as CaO, MgO, and mixed metal oxides have shown promising catalytic activity due to their strong basic properties and relatively low cost. [12]

Recent research has focused on improving catalyst performance through modification and support materials. The incorporation of transition metal oxides and porous carbon supports can enhance catalyst surface area, active site dispersion, and resistance to deactivation. Such improvements are particularly important for achieving high biodiesel yields under mild operating conditions while maintaining catalyst stability during repeated reaction cycles. Consequently, the development of advanced heterogeneous catalysts remains an active area of research aimed at improving the efficiency and sustainability of biodiesel production. [13]

2.4 Heterogeneous Catalysts in Biodiesel Production

Heterogeneous catalysts are catalysts that exist in a different phase from the reactants, typically as solid materials interacting with liquid reactants. Unlike homogeneous catalysts, heterogeneous catalysts can be easily separated from the reaction mixture through filtration, centrifugation, or sedimentation. This characteristic significantly simplifies downstream processing and allows catalyst recovery and reuse in multiple reaction cycles. The ability to recycle catalysts reduces operating costs and minimizes waste generation, making heterogeneous catalysis an attractive option for sustainable biodiesel production. [14]

One of the major advantages of heterogeneous catalysts is their ability to reduce environmental impacts associated with biodiesel production. Homogeneous catalysts often require large volumes of water for product purification and catalyst neutralization, generating substantial amounts of wastewater. In contrast, heterogeneous catalysts eliminate many of these purification requirements because the solid catalyst can be separated directly from the reaction mixture. This reduces water consumption, lowers wastewater treatment costs, and contributes to a greener production process. Furthermore, heterogeneous catalysts are generally less corrosive than liquid acid and base catalysts, reducing equipment degradation and maintenance requirements.

The performance of a heterogeneous catalyst is strongly influenced by its physicochemical properties. Parameters such as surface area, pore volume, pore size distribution, active site density, and catalyst basicity or acidity directly affect catalytic activity and biodiesel yield. Catalysts with larger surface areas generally provide more accessible active sites, promoting greater interaction between reactants and catalyst surfaces. Similarly, appropriate pore structures facilitate mass transfer and improve the diffusion of reactants and products within the catalyst matrix. Therefore, catalyst design and preparation methods play an important role in determining biodiesel production efficiency. [14]

Despite their numerous advantages, heterogeneous catalysts also face several challenges. Compared with homogeneous catalysts, they often exhibit slower reaction kinetics due to mass transfer limitations between the solid catalyst and liquid reactants. Catalyst deactivation caused by leaching, poisoning, sintering, or deposition of reaction by-products can also reduce long-term performance. Furthermore, some heterogeneous catalysts require relatively high reaction temperatures and extended reaction times to achieve comparable biodiesel yields. These challenges have motivated

extensive research aimed at developing advanced catalyst systems with improved activity, stability, and reusability. [15]

Recent developments have focused on mixed metal oxides, nanostructured catalysts, and supported catalyst systems that combine high catalytic activity with excellent stability. In particular, carbon-supported metal oxide catalysts have shown considerable promise due to their enhanced surface area, improved active site dispersion, and resistance to catalyst deactivation. Such catalysts provide a pathway toward more efficient and sustainable biodiesel production processes. [16] The development of Ca-Fe oxide supported on activated carbon, which is the focus of the present study, represents one such strategy aimed at overcoming the limitations of conventional heterogeneous catalysts while maintaining high biodiesel conversion efficiency.

2.5 Metal Oxide Catalysts for Biodiesel Production

Metal oxide catalysts have attracted significant attention in biodiesel production due to their high catalytic activity, thermal stability, low cost, and ease of recovery. Among the various heterogeneous catalysts investigated, metal oxides are particularly promising because they possess acidic, basic, or amphoteric surface properties that can effectively catalyze transesterification and esterification reactions. Their solid nature allows easy separation from the reaction mixture and facilitates catalyst reuse, making them more environmentally friendly and economically viable than conventional homogeneous catalysts. As a result, metal oxide catalysts have become one of the most extensively studied catalyst groups for sustainable biodiesel production. [17]

The catalytic performance of metal oxides largely depends on their surface chemistry, crystal structure, pore characteristics, and active site distribution. During transesterification, basic metal oxides promote the formation of alkoxide ions from alcohol molecules, which subsequently attack triglyceride molecules to produce fatty acid methyl esters (FAME). Therefore, catalysts with strong basic sites generally exhibit higher activity in biodiesel production from low free fatty acid (FFA) feedstocks. In contrast, acidic metal oxides are more suitable for feedstocks containing high FFA content because they can simultaneously catalyze esterification and transesterification reactions. [18]

Among alkaline earth metal oxides, calcium oxide (CaO) is one of the most widely investigated catalysts for biodiesel production. CaO possesses strong basicity, low toxicity, high catalytic activity, and relatively low cost. Additionally, calcium-based materials can be obtained from natural resources

and waste-derived materials such as eggshells, seashells, and limestone, making them attractive from both economic and environmental perspectives. Numerous studies have reported biodiesel yields exceeding 90% when CaO is used under optimized reaction conditions. However, CaO suffers from several drawbacks, including catalyst leaching, sensitivity to atmospheric moisture and carbon dioxide, and gradual deactivation during repeated use. These limitations can negatively affect catalyst stability and long-term performance. [19]

Transition metal oxides have also been extensively studied for biodiesel production. Iron oxide (Fe_3O_4), titanium dioxide (TiO_2), zinc oxide (ZnO), and zirconium oxide (ZrO_2) are among the most commonly investigated materials. Iron oxide is particularly attractive because of its low cost, chemical stability, and magnetic properties, which facilitate catalyst separation using an external magnetic field. Although Fe_3O_4 alone generally exhibits lower catalytic activity than strongly basic catalysts such as CaO, it can improve catalyst stability and structural integrity when incorporated into mixed metal oxide systems. The magnetic nature of Fe_3O_4 also simplifies catalyst recovery and reuse, making it an important component in advanced biodiesel catalysts. [20]

To overcome the limitations associated with single metal oxides, researchers have increasingly focused on mixed metal oxide catalysts. Combining two or more metal oxides can enhance catalytic activity, increase surface area, improve active site distribution, and reduce catalyst deactivation. Mixed oxide catalysts often exhibit synergistic effects resulting from interactions between different metal species. For example, CaO- Fe_3O_4 catalysts combine the strong basicity of calcium oxide with the structural stability and magnetic properties of iron oxide. Such combinations can improve catalyst durability while maintaining high biodiesel conversion efficiency. Recent studies have demonstrated that mixed metal oxide catalysts frequently outperform their individual components due to enhanced physicochemical properties and improved resistance to leaching.

Given these considerations, CaO- Fe_3O_4 -based catalyst systems have emerged as promising candidates for biodiesel production. The combination of high basicity, improved stability, magnetic recoverability, and potential support interactions offers opportunities to develop efficient and reusable heterogeneous catalysts. Consequently, research on carbon-supported CaO- Fe_3O_4 catalysts continues to expand as a strategy for improving biodiesel production efficiency and sustainability.

2.6 Carbon-Supported Catalysts

Carbon-supported catalysts have gained significant attention in biodiesel production due to their high surface area, porosity, thermal stability, and ability to enhance dispersion of active catalytic species. In heterogeneous catalysis, support materials play a crucial role in improving catalyst performance by increasing the accessibility of active sites, reducing particle agglomeration, and enhancing resistance to sintering and leaching. Carbon-based materials are particularly attractive because they are chemically stable under reaction conditions, relatively low-cost, and can be derived from both natural and waste biomass sources, making them environmentally sustainable options for catalyst design. [21]

In biodiesel production, carbon supports are commonly used to anchor metal or metal oxide active phases such as CaO , Fe_3O_4 , ZnO , and mixed oxides. The interaction between the active metal species and the carbon surface enhances catalyst stability and improves catalytic efficiency by preventing sintering during high-temperature calcination and repeated reaction cycles. Additionally, the porous structure of carbon materials facilitates improved mass transfer between the immiscible reactants (oil and alcohol), thereby enhancing reaction rates and overall biodiesel yield. [22]

Carbon-based catalyst supports are derived from various sources and are generally classified into activated carbon, biochar, and engineered carbon nanostructures. Activated carbon is the most widely used due to its extremely high surface area (often exceeding $1000 \text{ m}^2/\text{g}$) and well-developed pore structure, which allows excellent dispersion of catalytic species. Biochar, on the other hand, is produced from biomass waste through pyrolysis and is considered a more sustainable and low-cost alternative. Although biochar generally has lower surface area than activated carbon, its properties can be significantly improved through activation and functionalization processes. [23]

Despite their advantages, carbon-supported catalysts also present certain challenges. These include pore blockage due to deposition of reaction by-products, structural changes under harsh reaction conditions, and variability in properties depending on precursor source and synthesis conditions. Additionally, catalyst performance is strongly influenced by parameters such as impregnation method, calcination temperature, and metal loading. Therefore, careful optimization of preparation conditions is essential to achieve high catalytic efficiency and long-term stability. [23]

In the context of this study, carbon-supported Ca-Fe oxide catalysts are particularly promising due to synergistic effects between the support and active phases. Activated carbon provides a high-surface-

area matrix for dispersion, CaO supplies strong basic active sites for transesterification, and Fe₃O₄ enhances structural stability and enables magnetic separation. This combination improves catalytic efficiency, simplifies catalyst recovery, and enhances reusability, making it suitable for sustainable biodiesel production systems.

2.7 Critical Analysis of Previous Studies and Research Gap

Extensive research has been conducted on heterogeneous catalysts for biodiesel production, with a strong focus on improving catalytic activity, reusability, and environmental sustainability. Studies consistently show that heterogeneous catalysis offers significant advantages over homogeneous systems, particularly in terms of catalyst recovery, reduced wastewater generation, and improved process sustainability. However, despite these advantages, commercial-scale adoption remains limited due to unresolved challenges in catalyst stability, deactivation, and inconsistent performance across different feedstocks. [26]

A major portion of literature focuses on calcium oxide (CaO)-based catalysts, which are widely recognized for their strong basicity and high catalytic activity in transesterification reactions. CaO has been reported to achieve high biodiesel yields under optimized conditions, particularly when derived from low-cost waste materials such as eggshells and limestone. However, multiple studies highlight significant drawbacks including moisture sensitivity, carbonation by atmospheric CO₂, and leaching into the reaction medium, all of which contribute to rapid deactivation and reduced reusability.

Iron-based catalysts, particularly Fe₃O₄, have attracted attention due to their magnetic properties, chemical stability, and ease of separation using external magnetic fields. These features make Fe₃O₄ highly suitable for catalyst recovery and reuse in repeated reaction cycles. However, multiple studies indicate that Fe₃O₄ alone exhibits relatively low catalytic activity in transesterification, limiting its effectiveness as a standalone catalyst. As a result, Fe-based catalysts are often used as support materials or combined with more active oxides such as CaO to enhance performance while maintaining structural stability.

Carbon-based supports such as activated carbon and biochar have also been widely studied due to their high surface area, porosity, and surface chemistry. These materials improve catalyst performance by enhancing dispersion of active metal oxides and reducing agglomeration during synthesis and reaction. However, literature also reports challenges such as pore blockage due to

reaction by-products, structural degradation under high-temperature conditions, and variability in performance depending on precursor source and activation method. These inconsistencies limit reproducibility and scalability of carbon-supported catalyst systems.

Recent studies increasingly focus on mixed metal oxide catalysts to overcome the limitations of single-component systems. In particular, Ca-Fe-based catalysts have shown promising synergistic effects, where CaO provides strong basic active sites for transesterification, while Fe_3O_4 contributes magnetic separability and structural reinforcement. However, reported performance varies significantly across studies due to differences in synthesis methods, calcination temperatures, metal ratios, and support interactions. This indicates a lack of standardized preparation protocols and insufficient understanding of structure-activity relationships.

Overall, the literature clearly demonstrates that while heterogeneous catalysis has significantly advanced biodiesel production, important gaps remain in catalyst design, reproducibility, and long-term stability. In particular, Ca-Fe oxide supported on activated carbon remains underexplored despite its strong potential to integrate high basicity, magnetic recovery, and improved dispersion. This identifies a clear research gap and justifies the present study, which focuses on the synthesis, characterization, and performance evaluation of this catalyst system for biodiesel production.

3 Materials And Methodology

3.1 Introduction

The present study adopts a laboratory-based experimental approach focused on the development and evaluation of a heterogeneous catalyst for biodiesel production. All experimental activities are carried out in the Kinetic Laboratory of the Department of Chemical Engineering at Bangladesh University of Engineering and Technology (BUET). The research is designed to systematically synthesize catalyst materials, investigate their key properties, and evaluate their performance under controlled laboratory conditions. Experimental variables such as temperature, pH, reaction time, and reagent concentration are carefully monitored and controlled to ensure safe operation, reproducibility, and reliability of the results.

The experimental work consisted of four major stages: synthesis of Fe_3O_4 nanoparticles, preparation of $\text{CaO-Fe}_3\text{O}_4$ supported activated carbon catalysts, biodiesel production through transesterification of soybean oil, and characterization of both catalyst and biodiesel products. Catalyst characterization was performed using X-ray diffraction (XRD) analysis to determine the crystalline phases present in the synthesized catalyst. The produced biodiesel was evaluated through density, cloud point, and pour point measurements according to established standard methods.

The methodology presented in this final report addresses the limitations identified in the interim submission by refining the catalyst synthesis procedure, clearly defining experimental variables (oxide-to-carbon ratios), and incorporating standardized testing methods for biodiesel characterization.

3.2 Research Design

3.2.1 Experimental Framework

The present study followed an experimental research design involving catalyst synthesis, catalyst characterization, biodiesel production and fuel property evaluation. Initially, magnetite Fe_3O_4 nanoparticles were synthesized through a chemical co-precipitation method using ferrous sulfate heptahydrate and ferric nitrate nonahydrate as iron precursors. The synthesized Fe_3O_4 nanoparticles were subsequently combined with calcium nitrate and activated carbon support through a wet impregnation technique followed by thermal calcination under a nitrogen atmosphere.

Three catalyst compositions containing different carbon to oxide ratios (70:30, 80:20, and 90:10) were prepared to investigate the effect of catalyst composition on biodiesel production performance. The synthesized catalysts were then employed in the transesterification of soybean oil using methanol

under controlled reaction conditions. Following the reaction, the catalyst was separated and the biodiesel product was recovered and characterized.

3.2.2 Methodological Flowchart

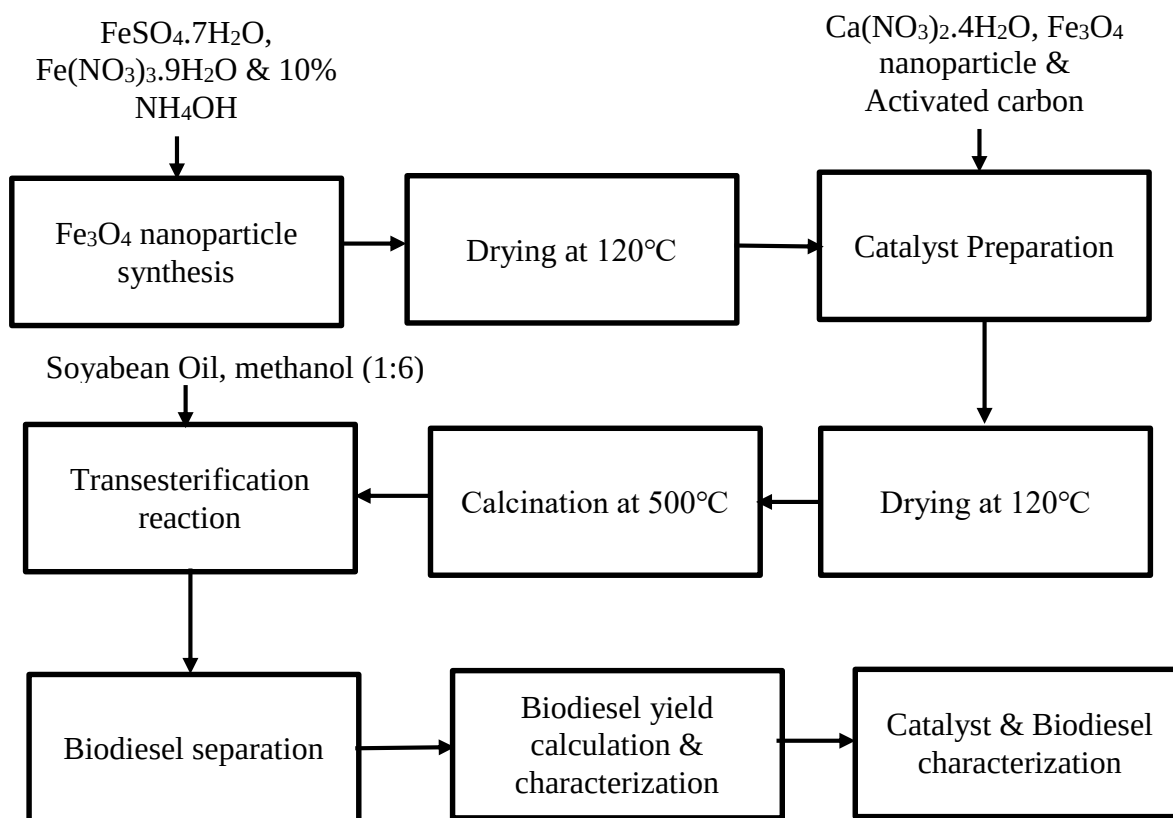


Figure 4: Flow diagram of the overall experimental methodology.

The overall experimental methodology consisted of several sequential stages, beginning with the synthesis of Fe_3O_4 nanoparticles through a co-precipitation process using ferrous sulfate, ferric nitrate, and ammonium hydroxide. The synthesized nanoparticles were dried and subsequently combined with calcium nitrate and activated carbon to prepare the catalyst precursor mixture. The resulting material was dried and calcined at 500°C under a nitrogen atmosphere to obtain the $\text{CaO-Fe}_3\text{O}_4/\text{activated carbon}$ catalyst. The prepared catalyst was then utilized in the transesterification of soybean oil with methanol to produce biodiesel. Following the reaction, the biodiesel was separated from the reaction mixture and the catalyst was recovered for further analysis. Finally, biodiesel yield was calculated, and both the catalyst and biodiesel products were characterized to evaluate catalyst performance and fuel quality.

3.3 Materials

3.3.1 Chemical & Reagents

Table 1: Different type of chemicals, formula and purpose used in the experiment

Chemical	Formula	Purpose
Ferrous sulfate heptahydrate	$\text{FeSO}_4 \cdot 7\text{H}_2\text{O}$	Fe_3O_4 precursor
Ferric nitrate nonahydrate	$\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$	Fe_3O_4 precursor
Calcium nitrate tetrahydrate	$\text{Ca}(\text{NO}_3)_2 \cdot 4\text{H}_2\text{O}$	Calcium source
Ammonium hydroxide	NH_4OH	Precipitating agent
Activated carbon	AC	Catalyst support
Methanol	CH_3OH	Alcohol reactant
Soybean Oil	-	Biodiesel feedstock
Distilled water	H_2O	Solvent

3.4 Catalyst Synthesis

3.4.1 Synthesis of Fe_3O_4 nanoparticles

Equipments and Instruments Used :_Burette, Clamp, Stand, Magnetic Hot Plate Stirrer, Beaker.

Fe_3O_4 nanoparticles were synthesized through a co-precipitation technique, which is widely employed for magnetite preparation due to its simplicity and effectiveness.

Initially, 6.9 g of ferrous sulfate heptahydrate ($\text{FeSO}_4 \cdot 7\text{H}_2\text{O}$) was dissolved in deionized water to prepare 50 mL of solution under continuous stirring. Separately, 20.2 g of ferric nitrate nonahydrate [$\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$] was dissolved in distilled water to prepare 100 mL of solution as per the setup described in the figure 5.

The ferrous sulfate solution was stirred at room temperature using a magnetic stirrer operating at approximately 300-500 rpm. The ferric nitrate solution was then added intermittently to the ferrous solution while maintaining continuous agitation. To minimize oxidation by atmospheric oxygen, the reaction vessel was covered with parafilm.

The mixed solution was heated to approximately 70°C. A 10 wt.% ammonium hydroxide solution was prepared by diluting 12 mL of 25 wt.% NH_4OH with 18 mL of distilled water. The ammonium hydroxide solution was rapidly introduced into the reaction mixture using a burette. Upon addition of the precipitating agent, the solution color gradually changed from brownish to jet black, indicating the formation of Fe_3O_4 nanoparticles.

The resulting black precipitate was separated through filtration using Whatman No. 42 filter paper. The collected precipitate was subsequently dried in an oven at approximately 120°C overnight. After drying, the solid material was ground using a mortar and pestle to obtain a fine Fe_3O_4 powder suitable for catalyst preparation.

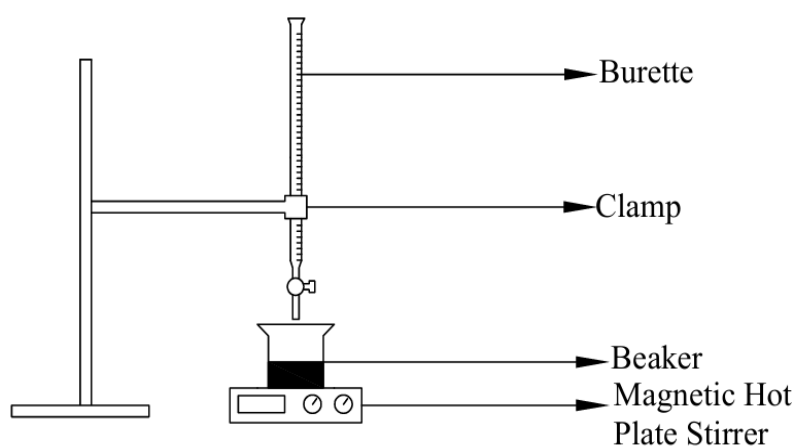


Figure 5: Schematic diagram of the Fe_3O_4 nanoparticle synthesis setup.

This setup was used for the synthesis of Fe_3O_4 nanoparticles through the controlled addition of precipitating agent to the iron salt solution. Continuous stirring and heating ensured homogeneous mixing and promoted the formation of magnetite nanoparticles with uniform properties.

3.4.2 Preparation of $\text{CaO-Fe}_3\text{O}_4$ Precursor Mixture

Equipments and Instruments Used : Burette, Clamp, Beaker and Magnetic Hot Plate Stirrer.

A mixed oxide precursor solution was prepared using Fe_3O_4 nanoparticles and calcium nitrate tetrahydrate. Approximately 0.55 g of synthesized Fe_3O_4 powder and 6.66 g of $\text{Ca}(\text{NO}_3)_2 \cdot 4\text{H}_2\text{O}$ were weighed to achieve an Fe molar ratio of 1:4. Both components were dissolved and dispersed in 40 mL of distilled water under continuous stirring to produce a homogeneous precursor suspension according to the figure 6.

The selected Fe-to-Ca ratio was intended to provide sufficient calcium oxide active sites while maintaining the magnetic properties contributed by the Fe_3O_4 phase. Similar mixed metal oxide systems have demonstrated promising catalytic activity in biodiesel production owing to their enhanced basicity and structural stability.

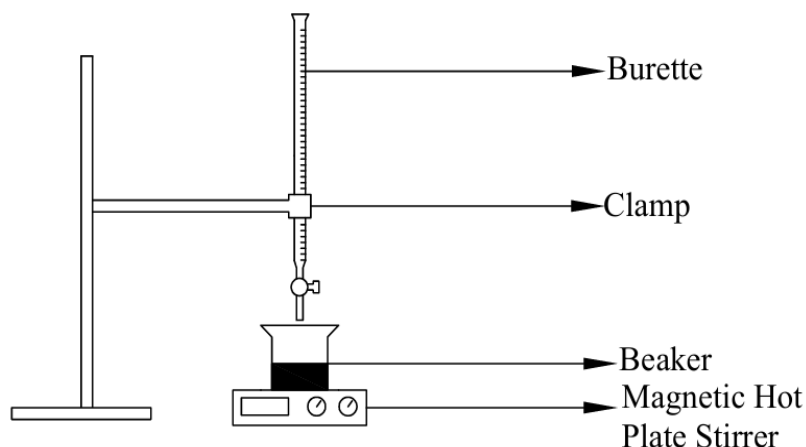


Figure 6: Schematic diagram of the $\text{CaO-Fe}_3\text{O}_4$ precursor mixture through wet impregnation.

This setup was used for the wet impregnation of activated carbon with calcium and iron precursor solutions. The precursor solution was added dropwise from a burette into a beaker containing activated carbon while being continuously stirred and heated using a magnetic hot plate stirrer to ensure uniform distribution of metal species on the support.

3.4.3 Catalyst Impregnation Procedure

Activated carbon was employed as the catalyst support due to its high surface area, porous structure and thermal stability. A predetermined amount of activated carbon was dispersed in distilled water to form a uniform suspension. The activated carbon suspension was subsequently added dropwise into the mixed Ca-Fe precursor solution while stirring continuously at approximately 500 rpm. The resulting slurry was dried at 120 °C overnight hours in a drying oven.

The wet impregnation technique was selected because it promotes uniform distribution of active metal species over the support surface and facilitates intimate contact between the active phase and support material.

Three catalyst compositions containing carbon-to-oxide ratios of 70:30, 80:20, and 90:10 were prepared to investigate the influence of support content on catalyst performance during biodiesel production.

3.4.4 Catalyst Calcination

Equipments and Instruments Used : Muffle Furnace, Plug Flow Reactor, N₂ cylinder

Nitrogen gas was supplied continuously at a flow rate of approximately 8 mL min⁻¹ to provide an inert atmosphere and minimize undesirable oxidation reactions during thermal treatment. The sample was heated gradually to 500°C over a period of approximately one hour and maintained at this temperature for three hours. After completion of the calcination period, the furnace was allowed to cool naturally to room temperature under continuous nitrogen flow.

The thermal decomposition of calcium nitrate during calcination produced calcium oxide species that were dispersed over the Fe₃O₄-activated carbon support. The final product obtained after calcination as per the system in described in figure 7, was designated as CaO-Fe₃O₄/AC heterogeneous catalyst and subsequently utilized for biodiesel production experiments.

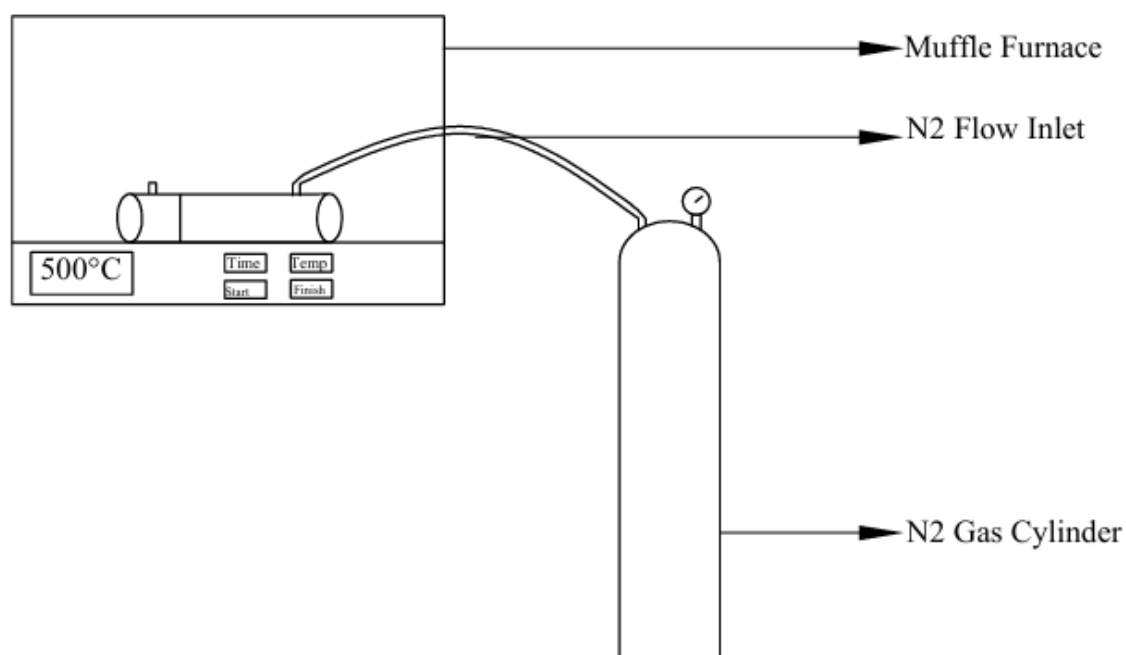


Figure 7: Schematic diagram of the calcination setup.

This setup illustrates the calcination process carried out in a muffle furnace under a continuous nitrogen atmosphere. The inert nitrogen flow prevented unwanted oxidation during thermal treatment, facilitating the formation of the active Ca-Fe oxide catalyst supported on activated carbon.

3.5 Catalyst Composition Variation

To investigate the influence of support content on catalytic performance, three different catalyst compositions were prepared by varying the oxide-to-carbon ratio. The ratios selected for this study were 70:30, 80:20, and 90:10 (CaO-Fe₃O₄: activated carbon). These variations were designed to evaluate how the amount of carbon support affects dispersion of active sites, surface accessibility and overall catalytic efficiency during biodiesel production.

Activated carbon plays a significant role in heterogeneous catalysis due to its high surface area, porosity, and adsorption capacity, which can enhance the accessibility of reactants to active sites. On the other hand, excessive reduction of active metal oxides may decrease catalytic activity. Therefore, optimizing the ratio between active phase and support is essential for achieving improved catalytic performance.

Each catalyst formulation was synthesized following the same impregnation and calcination procedure.

3.6 Biodiesel Production Experiment

3.6.1 Experimental Setup

Equipments and Instruments Used : Condenser, Flat Bottom Flask, Clamp, Magnetic Hot Plate Stirrer, Stand, Pipe.

The transesterification reaction was carried out in a laboratory-scale batch reactor consisting of a flat-bottom flask equipped with a reflux condenser, magnetic stirrer and heating mantle. This setup was used to ensure controlled temperature conditions and prevent methanol loss due to evaporation.

3.6.2 Transesterification Procedure

Soybean oil was used as the primary feedstock for biodiesel production. A known quantity of oil was introduced into the reactor, followed by methanol and the prepared heterogeneous catalyst. The mixture was initially stirred to ensure proper dispersion of the catalyst and alcohol within the oil phase.

The transesterification reaction proceeded under continuous stirring at elevated temperature (65°C). After completion of the reaction, the mixture was cooled to room temperature, and the catalyst was separated by filtration.

The reaction follows the general form:



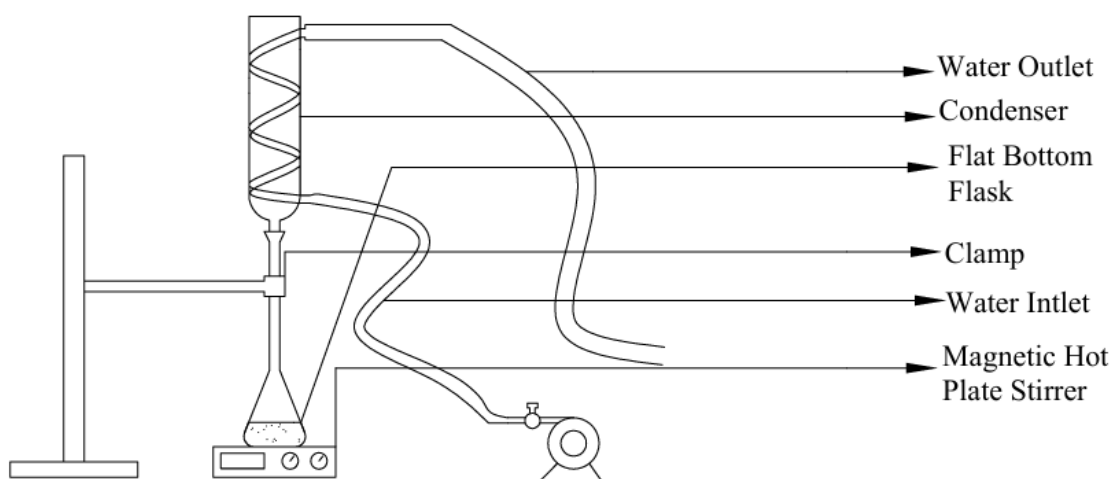


Figure 8: Schematic diagram of the transesterification reaction setup.

This setup was used for the transesterification of soybean oil with methanol in the presence of the synthesized catalyst. The reaction mixture was heated and continuously stirred using a magnetic hot plate stirrer, while the condenser minimized methanol loss by condensing and returning vaporized methanol to the reaction flask.

3.6.3 Catalyst Separation and Product Recovery

Equipments and Instruments Used :_Filter paper, Vacuum Pump.

After completion of the reaction, the solid catalyst was removed using filtration. The resulting liquid mixture containing biodiesel and glycerol was transferred to centrifuge tubes and centrifuged at 4500 rpm for 20 minutes to facilitate phase separation. The upper biodiesel layer was carefully collected for further characterization.

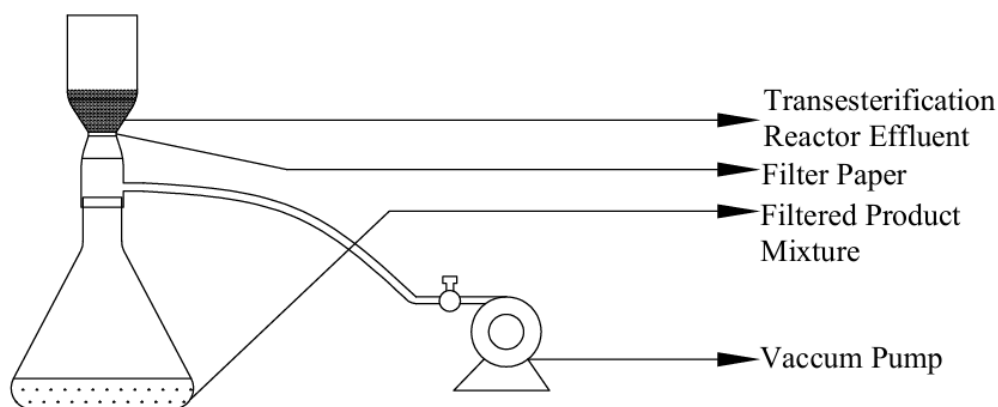


Figure 9: Schematic diagram of the filtration setup for recovery of catalyst & product.

This vacuum filtration apparatus was used to separate the solid catalyst from the reaction mixture after synthesis or washing steps. The application of vacuum accelerated filtration and enabled efficient recovery of the catalyst while removing excess liquid and impurities.

3.7 Catalyst Characterization

3.7.1 X-Ray Diffraction (XRD) Analysis

X-ray diffraction (XRD) analysis was employed to determine the crystalline phases, structural properties, and phase purity of the synthesized CaO-Fe₃O₄/activated carbon catalyst. XRD is a widely used technique in materials characterization as it provides information regarding interplanar spacing, crystal structure and identification of unknown phases by comparing diffraction patterns with standard reference databases

The crystalline phases of the synthesized catalyst were analyzed using X-ray diffraction (XRD). The analysis was carried out using Cu K α radiation generated at 40 kV and 50 mA. The diffraction patterns were recorded over a 2θ range of 5° to 80° with a step size of 0.01° and a scan speed of 35°/min in θ - 2θ scanning mode.

The raw XRD data were initially processed using the instrument's acquisition software and subsequently exported for detailed analysis. Phase identification was performed using X'PertHighScore Plus software, which compares experimental diffraction peaks with the International Centre for Diffraction Data (ICDD) database. The software enables qualitative phase analysis by matching peak positions (2θ values), relative intensities, and d-spacing values with standard reference patterns. This allowed identification of key crystalline phases such as Fe₃O₄ and CaO within the synthesized catalyst.

In addition to phase identification, MATCH! software was also used as a secondary validation tool for peak matching and phase confirmation. MATCH! provides an alternative database-driven approach to verify crystalline phases by comparing experimental diffraction patterns with standard reference files. The use of two independent software packages ensured higher reliability and reduced the possibility of misidentification of overlapping peaks, particularly in multiphase systems such as CaO-Fe₃O₄ supported on activated carbon.

For further quantitative and graphical analysis, the processed diffraction data were exported to Origin software (OriginLab). Origin was used for baseline correction, peak fitting, smoothing and graphical representation of XRD patterns. Peak deconvolution was performed where necessary to better distinguish overlapping diffraction peaks, and the final figures were prepared for inclusion in the thesis.

3.8 Biodiesel Characterization

3.8.1 Biodiesel Yield Calculation

The biodiesel yield was calculated using the following expression:

$$\text{Yield} = \frac{\text{Mass of biodiesel}}{\text{Theoretical mass of biodiesel}} \times 100$$

This method provides a direct measure of conversion efficiency and is commonly used in biodiesel production studies.

3.8.2 Density Measurement (ASTM D941)

Density is an important physical property of biodiesel that affects fuel injection, atomization, combustion efficiency, and overall engine performance. It is also used as an indicator of fuel purity and quality. Biodiesel with density values outside the recommended range may result in incomplete combustion, increased emissions, and poor engine operation. The density of the produced biodiesel was determined following ASTM D941 procedures. Before measurement, the biodiesel samples were allowed to reach a stable temperature under laboratory conditions. A clean and calibrated pycnometer was used for the analysis. The mass and volume of the biodiesel sample were measured. The measured density values were then corrected and interpreted according to ASTM D941 density–temperature relationships where applicable. The final results were compared with the EN 14214 biodiesel specification, which recommends a density range of 860-900 kg/m³ at standard conditions.

Density measurement provides useful information regarding fuel composition and conversion efficiency. Significant deviations from the standard range may indicate the presence of impurities, residual glycerol, unreacted oil, or incomplete transesterification.

3.8.3 Cloud Point (ASTM D2500)

The cloud point is the temperature at which the first visible wax crystals appear when a fuel is cooled under controlled conditions. It is an important indicator of biodiesel performance in cold climates because wax crystal formation can block fuel filters and reduce fuel flow. The cloud point of the biodiesel samples was determined according to ASTM D2500. In this method, a biodiesel sample was placed in a suitable test jar and cooled at a controlled rate using a cooling bath. At specified temperature intervals, the sample was removed and visually inspected under adequate lighting conditions.

The temperature at which a cloudy appearance was first observed due to the formation of wax crystals was recorded as the cloud point. Care was taken to avoid disturbing the sample during inspection, as agitation can affect crystal formation and lead to inaccurate observations.

The cloud point values obtained from this test were used to evaluate the low-temperature operability of the produced biodiesel and to compare the performance of biodiesel samples produced using different catalyst compositions.

3.8.4 Pour Point (ASTM D97)

The pour point is defined as the lowest temperature at which a fuel remains capable of flowing under prescribed testing conditions. It is a critical parameter for assessing biodiesel storage, transportation, and usability in cold weather environments. The pour point was measured according to ASTM D97. In this procedure, a biodiesel sample was first heated, if necessary, to remove any thermal history and then transferred into a standard test jar. The sample was subsequently cooled in a series of controlled cooling baths at specified temperature intervals. At every 3 °C decrease in temperature, the test jar was removed and tilted horizontally to observe whether the biodiesel still exhibited movement. This process was continued until no movement of the sample was detected within the specified observation period. The pour point was reported as the temperature that was 3 °C above the temperature at which the biodiesel ceased to flow. Lower pour point values indicate better cold-flow characteristics and improved suitability for operation under low-temperature conditions.

The measured pour point values were compared with biodiesel quality requirements and used to evaluate the effect of catalyst composition on the cold-flow properties of the produced biodiesel.

3.8.5 CHNS Elemental Analysis

Elemental analysis of the produced biodiesel samples was carried out to determine the carbon (C), hydrogen (H), nitrogen (N), sulfur (S) and oxygen (O) contents. The elemental composition provides important information regarding fuel quality, combustion characteristics, and energy content. Carbon and hydrogen are the primary contributors to fuel heating value, whereas oxygen content generally reduces the net calorific value due to its partially oxidized nature.

The biodiesel samples produced using catalysts with carbon-to-oxide ratios of 70:30, 80:20, and 90:10 were subjected to CHNS elemental analysis. The obtained elemental composition data were subsequently used for estimation of higher heating value (HHV) using an established empirical correlation. The measured elemental compositions were recorded and compared to evaluate the influence of catalyst composition on the chemical characteristics of the produced biodiesel.

3.8.6 Higher Heating Value (HHV) Determination

The higher heating value (HHV) of the produced biodiesel samples was estimated from the elemental composition obtained through CHNS analysis. HHV represents the total amount of heat released during complete combustion of a fuel and is one of the most important indicators of fuel energy content.

The HHV values were estimated using the empirical correlation proposed by Channiwala and Parikh (2002). This correlation is widely used in biodiesel and biofuel research because it predicts HHV from elemental composition data (C, H, O, N, S, and ash content) with good accuracy. Since elemental analysis was available for the produced biodiesel samples, the correlation provided a practical and reliable method for estimating energy content without the need for direct calorimetric measurements.[28]

$$\text{HHV, (MJ/kg)} = 0.3491C + 1.1783H + 0.1005S - 0.1034O - 0.0151N - 0.0211\text{Ash}$$

where:

- C = carbon content (wt%)
- H = hydrogen content (wt%)
- O = oxygen content (wt%)
- S = sulfur content (wt%)
- N = nitrogen content (wt%)

➤ Ash = ash content (wt%)

The calculated HHV values were used to compare the energy content of biodiesel produced using different catalyst compositions. Higher HHV values indicate greater energy release during combustion and are generally desirable for fuel applications.

3.9 Data Analysis

The experimental data obtained from catalyst synthesis, biodiesel production, and fuel property characterization were systematically analyzed to evaluate the performance of the developed CaO-Fe₃O₄/activated carbon catalyst. The analysis focused on comparing the effect of different catalyst compositions (70:30, 80:20, and 90:10 carbon-to-oxide ratios) on biodiesel yield and physicochemical properties, including density, cloud point and pour point.

Biodiesel yield was calculated based on the mass of biodiesel obtained relative to the mass of soybean oil used in the transesterification reaction. The yield values were used as the primary indicator of catalytic performance. The effect of catalyst composition on yield was evaluated through comparative trend analysis, where the performance of each catalyst formulation was ranked based on conversion efficiency.

In addition to yield analysis, the fuel properties of the produced biodiesel were compared against international standards, specifically EN 14214. Density values were evaluated against the acceptable range of 860-900 kg/m³, while cloud point and pour point values were assessed to determine cold-flow behavior suitability for practical applications. Deviations from standard values were interpreted in terms of possible incomplete transesterification, residual impurities or variations in fatty acid methyl ester composition.

No statistical modeling beyond descriptive comparison was performed; however, consistency between experimental runs was ensured through controlled reaction conditions and standardized measurement procedures.

4 Result and Discussion

4.1 Introduction

This chapter presents and discusses the experimental results obtained from the synthesis, characterization, and application of the Ca-Fe oxide supported on activated carbon catalyst for biodiesel production. The catalyst was synthesized through wet impregnation followed by calcination under a nitrogen atmosphere and subsequently characterized to evaluate its structural and surface properties.

The characterization results, including X-ray Diffraction (XRD), was used to investigate the crystalline phases, surface area, and pore characteristics of the prepared catalyst. These properties are important in determining the catalytic activity and effectiveness of the catalyst during the transesterification process.

The performance of the catalyst was evaluated through biodiesel production experiments using soybean oil as the feedstock. The effects of catalyst loading on biodiesel yield were investigated, and the quality of the produced biodiesel was assessed through yield analysis and comparison with established fuel standards. The obtained results were further compared with findings reported in previous studies to evaluate the effectiveness of the developed catalyst.

In addition, the chapter discusses the relationship between catalyst properties and biodiesel production performance based on the conceptual framework of the study. The limitations encountered during the experimental work are identified and analyzed, followed by recommendations for improvement. Finally, the engineering, economic, and sustainability aspects associated with the developed catalyst system are discussed.

4.2 Catalyst Characterization Results

4.2.1 XRD Analysis

X-ray diffraction (XRD) analysis was carried out to investigate the crystalline phases present in the synthesized $\text{CaO-Fe}_3\text{O}_4/\text{AC}$ catalyst before and after the transesterification reaction. The diffraction patterns were processed using HighScore Plus software & Match through background subtraction, smoothing and search-match analysis. The obtained patterns were used to identify the major crystalline phases and evaluate the structural stability of the catalyst during biodiesel production.

Fresh Catalyst Before Transesterification

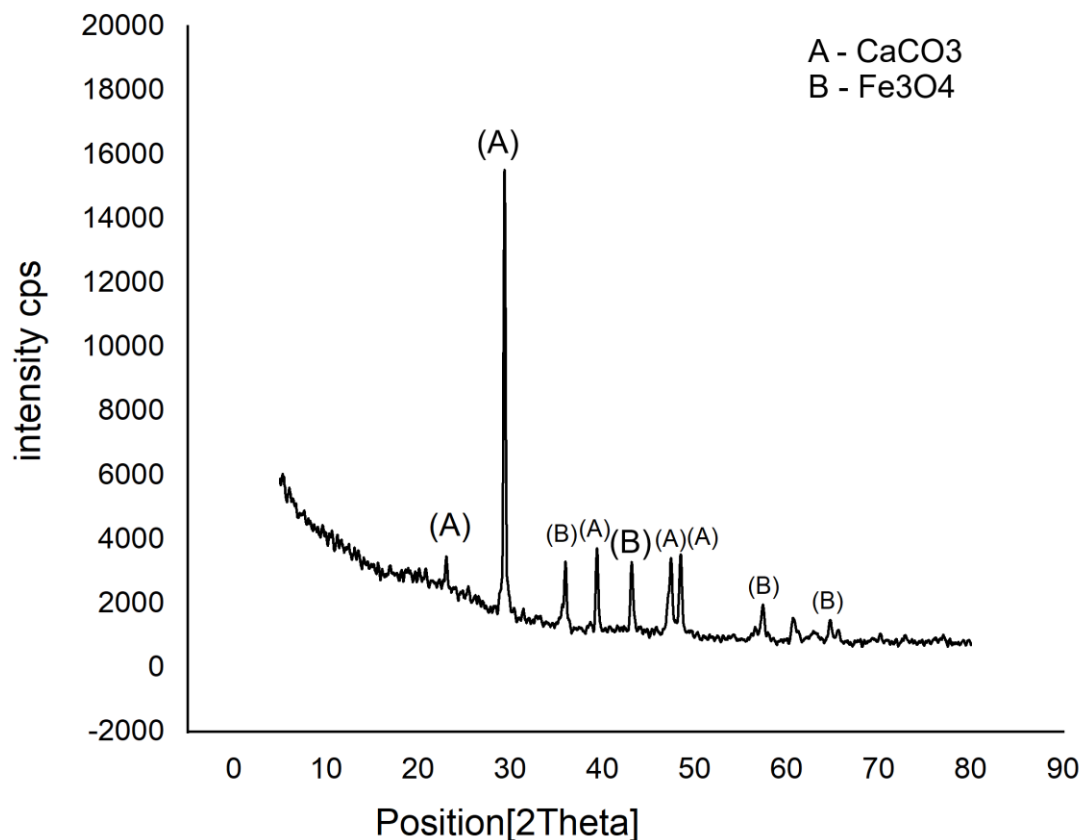


Figure 10: Pre-Transesterification Reaction peak identification

This figure presents the XRD pattern of the synthesized CaO-Fe₃O₄/AC catalyst prior to transesterification. Several diffraction peaks were observed at approximately 23.04°, 26.61°, 29.36°, 35.95°, 39.44°, 43.15°, 47.38°, 48.40°, 57.35°, 60.67°, and 64.73°, indicating the presence of crystalline phases within the catalyst matrix.

The most intense diffraction peak appeared at approximately 29.36°, suggesting that the corresponding phase is the dominant crystalline component in the catalyst. Search-match analysis performed using HighScore Plus identified this phase as calcium carbonate (CaCO₃).

Additional peaks observed at approximately 39.44°, 43.15°, 47.38°, 48.40°, and 57.35° also showed good agreement with the reference pattern of calcite, confirming the presence of calcium carbonate in the catalyst.

Search-match analysis also indicated the presence of an iron oxide phase. This result is consistent with the synthesis procedure, where Fe_3O_4 nanoparticles were incorporated into the catalyst structure. Characteristic diffraction peaks around 35.95° , 43.15° , 57.35° , and 64.73° are associated with iron oxide materials. Furthermore, the catalyst exhibited strong magnetic behavior during handling and separation, providing additional evidence for the presence of magnetite-related iron oxide phases within the catalyst. The broad peak observed around 23° may also be associated with the activated carbon support, which typically exhibits low crystallinity and broad diffraction features. The XRD results therefore confirm the successful incorporation of calcium-containing species and iron oxide phases onto the activated carbon support before transesterification.

Spent Catalyst After Transesterification

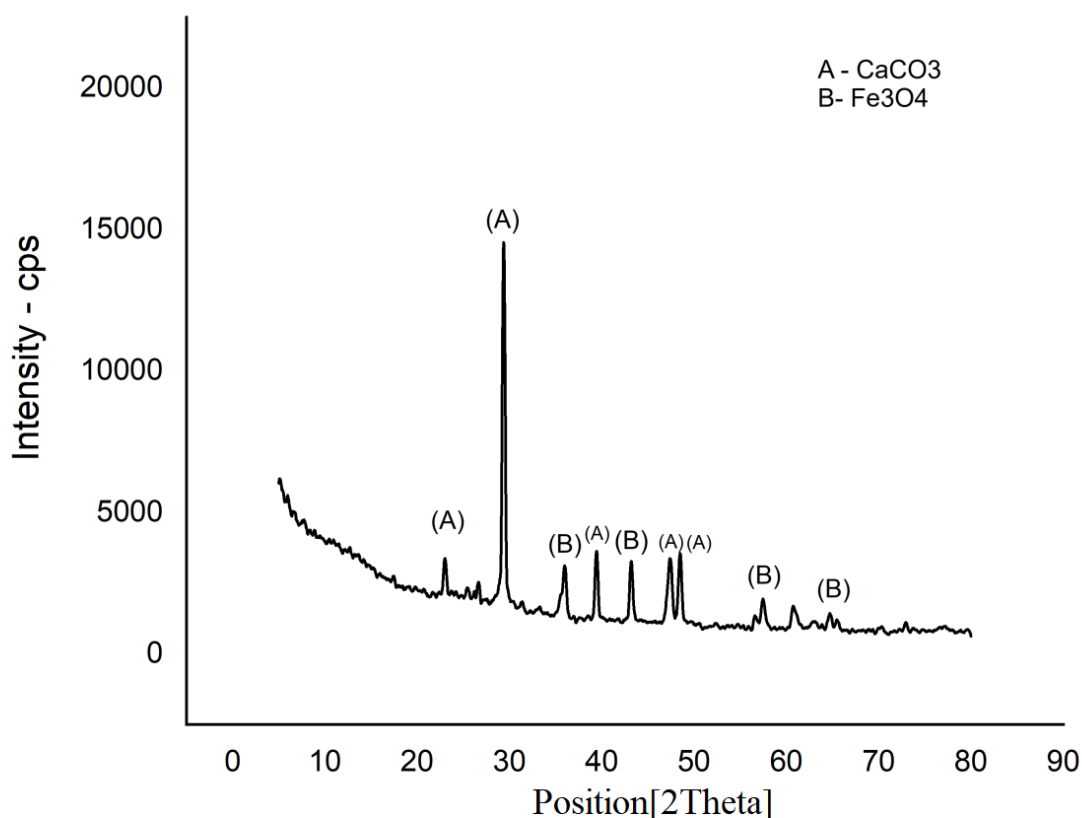


Figure 11: Post-Transesterification Reaction peak identification

The XRD pattern of the catalyst recovered after the transesterification reaction is shown in this figure. Similar diffraction peaks were observed at approximately 23.04° , 29.36° , 35.94° , 39.36° , 43.17° ,

47.43°, 48.51°, 57.32°, 60.65°, and 64.65°, indicating that the overall crystalline structure of the catalyst remained largely unchanged after biodiesel production.

Search-match analysis of the spent catalyst again identified calcium carbonate (CaCO_3) and an iron oxide phase as the dominant crystalline components. The persistence of the intense diffraction peak at approximately 29.36° and the corresponding calcite peaks suggests that calcium carbonate remained stable throughout the reaction process. Similarly, the iron oxide-related peaks remained present with only negligible shifts in peak position, indicating that the magnetic component of the catalyst retained its structural integrity during transesterification.

A comparison of the fresh and spent catalyst patterns revealed only minor variations in peak intensity and peak broadening. The positions of the major diffraction peaks remained essentially unchanged, suggesting that no significant phase transformation or crystalline degradation occurred during the reaction. The decreased intensity might be due to some form of leaching of the $\text{CaO-Fe}_3\text{O}_4$ from the catalyst support to the reaction product. A few weak additional reflections observed in the spent catalyst may be associated with adsorbed reaction products, surface carbonate species or calcium-containing intermediates formed during transesterification. However, these peaks were relatively low in intensity and did not significantly alter the overall diffraction pattern.

The preservation of the iron oxide diffraction peaks is particularly important because it indicates that the magnetic recovery functionality of the catalyst was maintained after use. The continued magnetic response of the catalyst following reaction further supports this conclusion.

Comparison of Fresh and Spent Catalysts

Comparison of the XRD patterns before and after transesterification demonstrates that the catalyst possesses good structural stability under the reaction conditions employed. The major diffraction peaks associated with calcium carbonate and iron oxide phases remained present after reaction, while only slight changes in peak intensity were observed.

The absence of significant peak shifts, disappearance of major peaks, or formation of substantial new crystalline phases indicates that the catalyst framework remained largely intact throughout the biodiesel production process. These findings suggest that the $\text{CaO-Fe}_3\text{O}_4/\text{AC}$ catalyst exhibits satisfactory crystallographic stability and resistance to structural degradation during transesterification.

4.2.2 Catalyst Yield Analysis

During catalyst preparation, a target batch mass of 10 g was initially set for each activated carbon-oxide composition (90:10, 80:20, and 70:30). However, due to unwanted material loss during calcination, the actual recovered catalyst mass was lower than the expected value. The catalyst yield was therefore calculated based on the ratio of actual recovered mass to the intended 10 g batch size.

Table 2: Catalyst Yield Analysis

Composition	SI No	Achieved mass (g)
90-10	1	6.8
	2	7.62
	3	7.68
80-20	1	7.06
	2	3.33
	3	6.05
	4	4.58
70-30	1	1.51
	2	2.23
	3	1.62
	4	7.22

The results show that catalyst yield generally decreases and becomes more inconsistent as the oxide content increases. This change is mainly related to the physical nature of the catalyst formed during calcination. At higher oxide compositions, the calcination process causes stronger oxide formation and breakdown of precursor materials. This leads to a more porous, light, and loosely packed (fluffy) catalyst structure. Because of this, the bulk density of the material becomes low and the particles are not tightly bound.

Due to this loose and low-density structure, fine catalyst particles can easily be carried away by the flowing gas inside the reactor during calcination. This results in partial loss of solid material through the outlet, which reduces the recovered mass and therefore lowers the catalyst yield. This effect is more significant in samples with higher oxide content, where pressure builds up more and more prone to material loss.

One exception was observed in the 70:30 composition, where a higher yield of 72.2% was obtained in one run. This does not follow the general trend.

This improvement happened because a different reactor setup was used in that run. Instead of a plug flow system, a larger closed box-type reactor was used. This reduced gas flow impact inside the reactor and helped prevent small particles from escaping.

As a result, less material was lost during calcination, and more solid catalyst was recovered.

Overall, the differences in catalyst yield are mainly caused by the physical structure of the catalyst and the reactor conditions during calcination. When the catalyst becomes too light and fluffy, more material is lost during processing. Improving the reactor design and controlling gas flow can help reduce this loss and improve catalyst recovery in future work.

4.3 Biodiesel Production Results

The performance of the synthesized Ca-Fe oxide supported on activated carbon catalyst was evaluated through the transesterification of soybean oil using methanol as the alcohol source. The reaction was conducted under controlled operating conditions, including a methanol-to-oil molar ratio of 6:1, a reaction temperature of 60-65 °C, and a reaction time of 3 hr. The catalyst was first mixed with methanol to promote the formation of active methoxide species before being introduced into the soybean oil.

Following the completion of the transesterification reaction, the catalyst was separated by filtration and the reaction products were subjected to centrifugation to facilitate phase separation between biodiesel and glycerol. The resulting biodiesel was collected and analyzed to evaluate the effectiveness of the catalyst in promoting the conversion of triglycerides into fatty acid methyl esters (FAME).

The biodiesel production results were assessed based on the yield obtained under different catalyst activated carbon-oxide ratio conditions. The observed trends provide insight into the catalytic performance of the prepared catalyst and the influence of catalyst concentration on reaction efficiency. The results are discussed in detail in the following subsections.

4.3.1 Biodiesel yield analysis based on the effect of activated carbon-oxides ratio

The effect of the activated carbon-to-oxide ratio on biodiesel production was investigated using catalysts with different compositions of activated carbon and CaO-Fe₃O₄ oxides. The biodiesel yield obtained for each catalyst composition is presented in the following table.

For 100 gram of soybean oil sample used, we expected equal amount of biodiesel. So for each sample, the expected mass of biodiesel is equal to the amount of oil used. [29]

Table 3: Biodiesel Yield analysis

Composition	SL No	Expected Mass (g)	Achieved Mass (g)	Yield (%)
90-10	1	60	37.44	62.4
	2	80	47.4	59.25
80-20	1	60	48.72	81.2
	2	45	33.86	75.24
70-30	1	45	41.4	92
	2	45	39.96	88.8

The results show that the biodiesel yield varies with the activated carbon-oxide ratio of the catalyst. The highest yield was obtained using the catalyst containing 70% activated carbon and 30% CaO-Fe₃O₄ oxides, with values of 92% and 88.8%, resulting in an average yield of approximately 90.4%.

For the 80:20 catalyst, the biodiesel yield was moderate, ranging from 75.24% to 81.2%, giving an average yield of approximately 78.2%. In comparison, the 90:10 catalyst showed the lowest performance, with yields of 62.4% and 59.25%, averaging around 60.8%.

A clear trend is observed where biodiesel yield increases with increasing oxide content in the catalyst. This suggests that the presence of CaO-Fe₃O₄ plays an important role in enhancing catalytic activity during the transesterification process. CaO provides strong basic active sites that promote the conversion of triglycerides into fatty acid methyl esters, while Fe₃O₄ contributes to structural stability and improves catalyst handling. Activated carbon helps in dispersing the active phases and increasing surface contact between reactants and catalyst.

The variation in yield across different compositions indicates that an optimum balance between activated carbon and oxide phases is necessary to achieve high catalytic performance. Among the tested ratios, the 70:30 composition provided the most effective catalytic activity and the highest biodiesel yield.

Overall, these results confirm that catalyst composition has a significant influence on biodiesel production efficiency, and proper optimization of the activated carbon-oxide ratio is essential for maximizing conversion performance.

4.4 Biodiesel Quality Assessment

The quality of the produced biodiesel was evaluated based on its yield, higher heating value (HHV), cloud point, pour point, and density. These properties are important indicators of fuel performance and determine the suitability of biodiesel for practical applications. The fuel properties of biodiesel produced using catalysts with different activated carbon-oxide ratios were compared with standard biodiesel specifications where applicable.

The elemental composition of the biodiesel samples was analyzed and used to estimate the higher heating value using the correlation proposed by Channiwala and Parikh. In addition, cold-flow properties and density were experimentally determined according to standard testing methods. The obtained results provide insight into the influence of catalyst composition on biodiesel quality and fuel characteristics.

4.4.1 Fuel Properties Comparison with Standards

In this section, cloud point, pour point, and density of the produced bio-diesel sample according to standard testing procedures is discussed.

Test Methods

Property	Testing Method
Cloud Point	ASTM D2500
Pour Point	ASTM D97
Density	ASTM D941

Results

The standards are taken from EN14214.

Property	Standard Value	Experimental Value	
		Composition (Carbon-Oxide)	Value
Cloud Point	Typical Value around -1 °C	70-30	1 °C
		80-20	-4 °C
		90-10	-2 °C
Pour Point	Upper limit 0 °C	70-30	-6.5 °C
		80-20	-10 °C
		90-10	-9 °C
Density	Between 860 - 900 kg/m ³	70-30	904.74 kg/m ³
		80-20	904.78 kg/m ³
		90-10	907.89 kg/m ³

The cloud point of the biodiesel samples ranged from -4 °C to 1 °C. The lowest cloud point was observed for the 80:20 sample (-4 °C), indicating improved low-temperature operability. In contrast, the 70:30 sample exhibited a cloud point of 1 °C, which is slightly higher and may indicate earlier wax formation at lower temperatures.

The pour point values ranged from -10 °C to -6.5 °C, all of which are below the upper acceptable limit of 0 °C. The 80:20 sample again showed the best cold-flow performance with the lowest pour point (-10 °C), suggesting better resistance to solidification under low-temperature conditions.

The density of the biodiesel samples ranged from 904.74 to 907.89 kg/m³, which is slightly higher than the EN 14214 standard range (860-900 kg/m³). This deviation may be attributed to incomplete conversion of triglycerides, residual impurities such as glycerol, or the presence of heavier intermediate compounds in the final product.

Overall, the biodiesel samples exhibited satisfactory cold-flow properties but slightly elevated density values compared to international standards. Among the tested catalysts, the 70:30 composition delivered the highest yield, while the 80:20 composition provided the best cold-flow performance. This indicates a trade-off between production efficiency and certain fuel properties depending on catalyst composition.

4.4.2 Higher heating value analysis

Table 4: Observed data for CHNS analysis.

Sample	N%	C%	H%	S%	O%
70-30	.305	73.571	10.958	-	15.166
80-20	.203	73.162	11.078	-	15.557
90-10	.175	74.297	11.401	-	14.127

The higher heating value (HHV) of the biodiesel samples was estimated using the elemental composition and the empirical correlation proposed by Channiwala and Parikh:

$$\text{HHV}(\text{MJ/kg}) = 0.3491\text{C} + 1.1783\text{H} + 0.1005\text{S} - 0.1034\text{O} - 0.0151\text{N}$$

The calculated HHV values for the biodiesel samples are presented below:

Table 5: Calculated data for HHV from CHNS data

Catalyst Composition	HHV (MJ/Kg)
70-30	37.02
80-20	36.98
90-10	37.91

The results show that the HHV values remain within a relatively narrow range (36.98-37.91 MJ/kg), indicating that catalyst composition has only a minor influence on the energy content of the final biodiesel product.

The highest HHV was observed for the biodiesel produced using the 90:10 catalyst (37.91 MJ/kg), which may be associated with its slightly higher carbon and hydrogen content and lower oxygen content. Since HHV increases with higher carbon and hydrogen fractions and decreases with oxygen content, small variations in elemental composition can lead to noticeable differences in energy content.

4.5 Comparison with Previous Studies

This section compares the performance of the developed Ca-Fe oxide supported on activated carbon catalyst with previously reported heterogeneous catalysts used for biodiesel production. The

comparison is based mainly on biodiesel yield and catalyst stability, which are two of the most critical performance indicators in transesterification reactions. The purpose of this comparison is to evaluate the effectiveness of the developed catalyst system and identify its relative advantages and limitations in relation to existing studies.

Overall, literature reports show that biodiesel production using heterogeneous catalysts typically achieves yields ranging from 70% to 95%, depending on catalyst type, reaction conditions, and feedstock quality. Metal oxide catalysts such as CaO, MgO, and mixed oxide systems often demonstrate high initial activity but vary significantly in stability and reusability. Carbon-supported and modified catalysts, on the other hand, tend to show improved stability but sometimes lower catalytic activity.

4.5.1 Comparison of Yield

The biodiesel yield obtained in this study using the Ca-Fe oxide supported on activated carbon catalyst reached a maximum average value of approximately 90.4% for the 70:30 activated carbon-oxide ratio (based on experimental runs of 92% and 88.8%). This yield is comparable to, and in some cases slightly lower than, values reported in similar studies using heterogeneous catalysts.

For instance, CaO-based catalysts derived from waste materials such as eggshells and shells typically report biodiesel yields in the range of 85% to 95% under optimized conditions. However, these catalysts often require strict reaction conditions and may suffer from performance decline upon reuse due to deactivation effects. Similarly, Fe-based catalysts generally show lower yields unless modified or combined with other active phases.

Mixed metal oxide catalysts reported in literature, particularly CaO-based composites, often achieve yields between 88% and 96%, depending on synthesis method and support material. In comparison, the approximately 90.4% yield obtained in this study demonstrates that the developed catalyst performs competitively within the mid-to-upper range of reported values.

The observed performance can be attributed to the synergistic effect between CaO and Fe₃O₄, where CaO provides strong basic active sites while Fe₃O₄ enhances dispersion and structural stability. Additionally, the activated carbon support contributes to improved surface area and reactant accessibility, which further supports catalytic efficiency.

4.5.2 Correlation with Conceptual Framework

This section establishes the relationship between the conceptual framework of the study and the experimental outcomes obtained during catalyst synthesis and biodiesel production. The conceptual framework of this research is based on the principle that catalyst composition and structure directly influence catalytic activity, which in turn determines biodiesel yield and fuel quality.

In this study, the independent variable is the catalyst composition, specifically the activated carbon-oxide ratio (CaO-Fe₃O₄ supported on activated carbon). The dependent variables include biodiesel yield and fuel properties such as higher heating value (HHV), density, cloud point, and pour point. The intermediate mechanism linking these variables is the catalytic effectiveness of active sites, surface area availability, and mass transfer efficiency during the transesterification reaction.

The experimental results clearly support this framework. The catalyst containing 70% activated carbon and 30% CaO-Fe₃O₄ oxides produced the highest biodiesel yield indicating that an optimal balance between active catalytic sites and support material enhances reaction efficiency. In contrast, increasing the proportion of activated carbon (80:20 and 90:10 ratios) reduced the number of available active oxide sites, leading to lower conversion efficiency. This confirms the conceptual assumption that catalytic activity is strongly dependent on active phase concentration and accessibility.

The role of activated carbon as a support material is also consistent with the framework. Activated carbon improves dispersion of CaO and Fe₃O₄ particles, increases surface area, and enhances reactant contact. However, excessive carbon content reduces the proportion of active catalytic sites, which negatively impacts biodiesel yield. This demonstrates a trade-off between surface enhancement and active site availability, which is a key concept in heterogeneous catalyst design.

Similarly, Fe₃O₄ contributes to structural stability and facilitates catalyst recovery due to its magnetic properties. Although it does not significantly increase catalytic activity, its presence improves catalyst usability and aligns with the framework's assumption that catalyst functionality is a combination of activity and recoverability.

The fuel property results further support the conceptual framework. While biodiesel yield showed strong dependence on catalyst composition, properties such as HHV remained relatively stable across samples, indicating that catalyst composition primarily influences conversion efficiency rather than intrinsic fuel energy content. However, variations in density and cold-flow properties suggest that incomplete conversion and residual impurities may still be influenced by catalyst effectiveness.

Overall, the experimental findings strongly validate the conceptual framework of the study, confirming that catalyst composition governs active site availability, which directly affects transesterification efficiency and biodiesel yield. The observed trends also highlight the importance of optimizing the balance between support material and active oxide phases to achieve both high catalytic performance and acceptable fuel properties.

4.6 Limitations of Findings

This section outlines the key limitations encountered during the experimental work and analysis of the Ca-Fe oxide supported on activated carbon catalyst for biodiesel production. These limitations are categorized into experimental, material, and analytical constraints that may have influenced the results and their interpretation.

4.6.1 Experimental Limitations

The experimental study was conducted at laboratory scale under controlled batch conditions. While this allows for systematic evaluation of catalyst performance, it does not fully represent industrial-scale biodiesel production systems. Factors such as continuous flow operation, large-scale mixing efficiency, and heat transfer effects were not considered in this work.

One of the main limitations was the incomplete transesterification reaction. Due to operational constraints such as fixed reaction time and limited optimization of reaction parameters (temperature, methanol-to-oil ratio, and mixing intensity), full conversion of triglycerides to fatty acid methyl esters (FAME) may not have been achieved. This could have contributed to slightly lower purity of biodiesel and variations in measured fuel properties.

Another important limitation was the absence of a washing and purification step after biodiesel separation. The biodiesel samples were analyzed directly after phase separation without water washing or additional purification. As a result, traces of impurities such as residual methanol, unreacted oil, glycerol, and catalyst particles may still have been present in the final product. This likely affected measured properties such as density and may have caused deviations from standard EN 14214 values

Furthermore, catalyst reusability and long-term cyclic stability were not extensively evaluated. As a result, the durability and performance retention of the catalyst over multiple reaction cycles could not be fully confirmed within the scope of this study.

In addition, the reaction was performed under a limited range of operating conditions, and only a few catalyst compositions (70:30, 80:20, and 90:10) were tested. This restricts the ability to identify the absolute optimum catalyst ratio or fully understand the effect of broader parameter variations such as reaction time, methanol-to-oil ratio, and temperature fluctuations.

4.6.2 Material Limitations

The catalyst preparation process relied on laboratory-grade chemical precursors and activated carbon as the support material. Variations in activated carbon properties, such as pore size distribution, surface functional groups, and ash content, may influence catalyst performance and reproducibility.

Additionally, the formation of $\text{CaO-Fe}_3\text{O}_4$ active phases depends strongly on calcination conditions. Slight variations in temperature control, heating rate, or nitrogen flow may affect particle size, phase purity, and dispersion of active species on the carbon surface. These factors can introduce inconsistencies in catalytic activity between different batches.

Another limitation is the potential partial loss of active CaO due to hydration or carbonation during handling and storage prior to reaction, which may reduce the effective catalytic activity.

4.6.3 Analytical Limitations

Although standard methods were used for biodiesel characterization (ASTM and EN-based procedures), some analytical limitations were present. The determination of biodiesel yield was based on experimental measurements that may include minor uncertainties due to separation efficiency and residual impurities in the biodiesel phase.

Elemental analysis used for HHV estimation relies on empirical correlations (Channiwala and Parikh correlation), which provide approximate values rather than direct experimental measurement of heating value. Therefore, the calculated HHV values may slightly deviate from actual calorific values.

Additionally, limited instrumentation was available for advanced catalyst characterization. Techniques such as TEM, XPS, and detailed surface chemistry analysis were not extensively performed, restricting deeper understanding of nanoscale structure and active site behavior.

Despite these constraints, the study successfully demonstrates the catalytic potential of Ca-Fe oxide supported on activated carbon for biodiesel production because XRD analysis catalyst stability and

70-30 ratio showed high (around 90% yield) showed. The identified limitations provide a basis for future improvements in experimental design, catalyst optimization, and analytical depth.

4.7 Mitigation Strategies and Improvements

This section presents recommended strategies to address the limitations identified in Section 4.9 and to improve the performance, reliability, and scalability of the Ca-Fe oxide supported on activated carbon catalyst system for biodiesel production.

4.7.1 Improvements in experimental design

To overcome limitations associated with laboratory-scale batch operation, future studies should consider conducting experiments under semi-pilot or continuous flow reactor conditions. This would provide more realistic insight into heat and mass transfer behavior, catalyst performance stability, and process scalability.

A wider range of operating parameters should also be investigated, including variations in reaction time, methanol-to-oil molar ratio, reaction temperature, and catalyst loading. This would allow for more precise optimization of biodiesel yield and better understanding of reaction kinetics. In addition, long-term catalyst reusability tests involving multiple reaction cycles (e.g., 5-10 cycles) should be performed to evaluate catalyst stability, deactivation rate, and regeneration potential.

4.7.2 Improvements in Catalyst Preparation

To improve catalyst reproducibility and performance, strict control of synthesis parameters such as calcination temperature, heating rate, and nitrogen flow rate should be maintained. These parameters directly influence the phase formation, particle size, and dispersion of CaO and Fe₃O₄ on the activated carbon support.

Surface modification of activated carbon (e.g., acid or base activation) may also be introduced to enhance functional groups and improve metal-support interaction. This can lead to better dispersion of active phases and improved catalytic efficiency.

Furthermore, optimizing the CaO to Fe₃O₄ ratio beyond the tested range could help identify a more precise optimum composition for maximizing both activity and stability.

4.7.3 Improvements in Analytical Methods

For more accurate evaluation of biodiesel quality, direct experimental determination of higher heating value using a bomb calorimeter is recommended instead of relying solely on empirical correlations. This would provide more precise energy content data.

Advanced characterization techniques such as Transmission Electron Microscopy (TEM), X-ray Photoelectron Spectroscopy (XPS), and Temperature-Programmed Desorption (TPD) should also be incorporated in future studies to better understand catalyst surface chemistry, active site distribution, and adsorption behavior.

Improved separation and purification techniques (e.g., advanced centrifugation or washing protocols) can also reduce impurities in biodiesel, leading to more accurate density and quality measurements. Implementing these mitigation strategies would significantly enhance catalyst performance, improve data reliability, and support the transition of this catalyst system from laboratory-scale research to potential industrial application in sustainable biodiesel production.

5 Impact and Implications

5.1 Introduction

With the exhaustion of fossil fuel reserves, increasing energy demand and growing concerns of environmental pollution, there is a growing need for sustainable and environmentally friendly energy sources. Due to its renewable, biodegradable and environmental friendly characteristics, biodiesel has proven to be an alternative fuel of interest. The synthesis of a Ca-Fe oxide catalyst supported on activated carbon was performed and used in transesterification of soybean oil to obtain biodiesel in the present study. The preparation and characterization of the catalyst and the production of biodiesel under optimized reaction conditions are proof of the potential of using heterogeneous catalytic systems for the production of renewable fuels.

In addition to the experimental results, it is necessary to consider the general implications of the catalyst developed and process for the production of biodiesel. This implies environmental, industrial, economic, occupational health and safety and sustainable aspects. All these factors can help to form a full picture of the value and impact of the current study and its application to sustainable energy development.

By utilizing magnetic Fe_3O_4 nanoparticles and structural calcium active sites on a porous carbon support, this catalytic matrix attempts to reconcile the historic trade-offs between homogeneous efficiency and heterogeneous process economics. The sections below analyze how this specific catalyst formula shapes the environmental balance, plant-scale economics, facility safety protocols and long-term sustainability metrics of an integrated biorefinery.

5.2 Environmental Impact

The primary environmental benefit of substituting a solid $\text{CaO-Fe}_3\text{O}_4/\text{AC}$ catalyst for standard homogeneous alternatives (such as liquid NaOH) lies in the drastic reduction of downstream wastewater pollution. Homogeneous systems demand extensive aqueous washing phases to extract dissolved soaps and residual alkalis from the crude fatty acid methyl esters (FAME) layer. This produces a highly alkaline, high-Chemical Oxygen Demand (COD) effluent that requires heavy chemical neutralization before discharge. In contrast, the heterogeneous $\text{CaO-Fe}_3\text{O}_4/\text{AC}$ matrix eliminates the washing stage entirely, minimizing the net volumetric water footprint of the purification block. The net carbon-neutral status of the biodiesel lifecycle is further preserved by utilizing an activated carbon carrier, which acts as a stable sink for carbonaceous biomass waste.

However, a critical environmental challenge identified during the phase stability and structural characterization of the catalyst is active site leaching. X-ray diffraction (XRD) data collected post-transesterification shows a notable reduction in active phase crystalline intensities, confirming that minor fractions of calcium and iron species migrate into the liquid phase.

One of the primary advantages of biodiesel production is its potential to reduce environmental pollution associated with fossil fuel consumption. Biodiesel is a renewable fuel derived from biological sources and generally produces lower emissions of carbon monoxide, particulate matter, sulfur oxides and unburned hydrocarbons during combustion compared to petroleum diesel. The use of biodiesel therefore contributes to improved air quality and reduced environmental degradation. The successful conversion of soybean oil into biodiesel demonstrates the possibility of replacing a portion of fossil-derived diesel with renewable fuel. Since the carbon dioxide released during biodiesel combustion is partially offset by the carbon absorbed during plant growth, biodiesel contributes to a lower net carbon footprint compared to conventional diesel fuels.

However, it should also be acknowledged that large-scale cultivation of soybean for biodiesel feedstock may create concerns related to land use, water consumption, and agricultural resource allocation. Therefore, future research should investigate the utilization of waste cooking oil and non-edible feedstocks to further improve the environmental performance of biodiesel production.

5.3 Economic and Industrial Implications

Transitioning from standard homogeneous transesterification to a heterogeneous ($\text{CaO-Fe}_3\text{O}_4/\text{AC}$) operational model introduces a distinct shift in capital expenditure (CAPEX) and operating expenditure (OPEX) structures. Homogeneous processing requires continuous, sacrificial catalyst chemical purchases since the active base is permanently lost in the product streams. By contrast, the $\text{CaO-Fe}_3\text{O}_4/\text{AC}$ catalyst based system capitalizes on structural magnetic properties driven by the co-precipitated Fe_3O_4 core, enabling simple magnetic separation. This rapid catalyst recovery loop significantly drives down long-term operating costs by preserving catalyst reusability over consecutive production cycles.

On a plant scale, integrating this heterogeneous pathway simplifies the downstream purification block by removing the acid-neutralization reactors and settling decanters needed for soap split phases. This compact layout directly balances the higher upfront costs of synthesizing composite nanostructured catalysts.

The catalyst developed in this work consists of relatively inexpensive and readily available materials such as calcium compounds, iron compounds and activated carbon. These materials are commercially

accessible and can be prepared using straightforward synthesis techniques, making scale-up potentially feasible.

The physical risks of catalyst clumping and filtration resistance against the savings of shorter reaction times. Left unmanaged, these issues can cause localized pressure drops and flow maldistributions in fixed-bed reactors, threatening continuous plant uptime.

5.4 Health and Safety Considerations

Health and safety are important considerations throughout biodiesel production and catalyst synthesis processes. Implementing the CaO-Fe₃O₄/AC preparation and reaction pathways requires a specialized Environmental Mitigation and Monitoring Plan (EMMP). The early catalyst synthesis steps- particularly the co-precipitation of Fe₃O₄ nanoparticles and subsequent impregnation onto fine activated carbon particles- present acute respiratory inhalation risks. Airborne nanostructured powders can easily bypass standard laboratory engineering controls requiring strict use of localized particulate filtration and respiratory protection to prevent pulmonary inflammation or long-term tissue accumulation.

The transesterification process involves the use of methanol, which is highly flammable and toxic. Exposure to methanol can cause serious health effects through inhalation, ingestion or skin contact. Because the heterogeneous reaction requires elevated temperatures near or above the boiling point of methanol (65°C), the system operates under pressure to keep the alcohol in the liquid phase. This combination of flammable vapor, pressure and fine carbonaceous dust creates a classic dust-explosion risk profile. Mitigation requires specialized industrial safety protocols.

Compared to homogeneous catalyst systems, heterogeneous catalysts generally reduce worker exposure to corrosive liquid catalysts and decrease the risk associated with catalyst handling during downstream processing. The easier separation of solid catalysts can also improve operational safety and reduce chemical waste generation.

From a fuel utilization perspective, biodiesel is considered less toxic and more biodegradable than conventional diesel fuel. In the event of accidental spills, biodiesel typically poses lower environmental and health risks due to its relatively rapid biodegradation and lower toxicity characteristics.

5.5 Sustainability Aspects

Sustainability represents one of the most significant motivations for biodiesel research. The catalyst system developed in this study aligns with several principles of sustainable chemical processing, including renewable resource utilization, waste minimization, and cleaner production technologies. The use of a heterogeneous catalyst supports sustainable manufacturing by reducing wastewater production and facilitating catalyst recovery. Furthermore, activated carbon-supported catalysts provide opportunities for catalyst regeneration and extended operational lifetimes, which can contribute to improved resource efficiency.

The engineered $\text{CaO-Fe}_3\text{O}_4/\text{AC}$ catalyst offers a compelling, environmentally conscious pathway for biodiesel synthesis by eliminating alkaline wastewater streams and leveraging magnetic separation to lower operating costs. However, these advantages must be carefully weighed against the realities of industrial operations. The economic savings of a re-usable catalyst can be quickly eroded by the costs of active site leaching, downstream ash contamination and the energy required to overcome phase mass-transfer limitations

Biodiesel produced from vegetable oils offers a renewable alternative to finite fossil fuel resources. The transition toward renewable fuels is essential for achieving long-term energy security and reducing greenhouse gas emissions. The successful production of biodiesel using the synthesized catalyst demonstrates the potential contribution of catalyst-assisted transesterification technologies to sustainable energy systems. Sustainability assessments must consider the entire life cycle of biodiesel production, including feedstock cultivation, transportation, processing and fuel utilization. Future investigations involving life-cycle assessment (LCA), catalyst reusability studies would provide a more comprehensive understanding of the sustainability performance of the developed process.

6 Conclusion and Recommendations

6.1 Introduction

This chapter presents the conclusions drawn from the study on Ca-Fe oxide supported on activated carbon as a heterogeneous catalyst for biodiesel production via transesterification of soybean oil. It summarizes the key findings of the research in relation to the stated objectives, including catalyst synthesis, characterization, and performance evaluation.

In addition, this chapter highlights the overall effectiveness of the developed catalyst system and its implications for biodiesel production. Based on the experimental outcomes, recommendations are provided for improving catalyst performance and process efficiency. Finally, potential directions for future research are discussed to support further development and optimization of the catalyst system for large-scale applications.

6.2 Summary of Key findings

The key findings of this study are summarized based on catalyst synthesis, performance evaluation, and biodiesel quality assessment.

The Ca-Fe oxide supported on activated carbon catalyst was successfully synthesized using wet impregnation followed by calcination under a nitrogen atmosphere. The combination of CaO, Fe₃O₄, and activated carbon provided a multifunctional catalyst system combining active basic sites, structural stability, and high surface area support characteristics.

Experimental results demonstrated that catalyst composition had a significant influence on biodiesel production. Among the tested configurations, the catalyst containing 70% activated carbon and 30% CaO-Fe₃O₄ oxides achieved the highest biodiesel yield of 92%. Lower oxide content (80:20 and 90:10 ratios) resulted in reduced catalytic activity and lower biodiesel yields of 80% and 60%, respectively.

Fuel property analysis showed that the produced biodiesel exhibited acceptable cold flow characteristics, with cloud point and pour point values generally within or near standard ranges. However, density values were slightly higher than EN 14214 standard limits, indicating possible traces of impurities or incomplete conversion.

The higher heating value (HHV) of the biodiesel samples, calculated using elemental composition, remained relatively consistent across all catalyst compositions, ranging from 36.98 to 37.91 MJ/kg.

This indicates that catalyst composition had minimal effect on the energy content of the produced biodiesel.

A comparison with previous studies showed that the developed catalyst performs competitively with reported heterogeneous catalysts, particularly CaO-based and mixed metal oxide systems, in terms of biodiesel yield. Additionally, the incorporation of Fe_3O_4 provided potential advantages in catalyst recovery due to its magnetic properties, while activated carbon improved dispersion and surface availability of active sites.

Overall, the findings confirm that the Ca-Fe oxide supported on activated carbon catalyst is an effective system for biodiesel production, with strong potential for optimization and further scale-up applications.

6.3 Conclusions

This study successfully demonstrated the synthesis and application of Ca-Fe oxide supported on activated carbon as a heterogeneous catalyst for biodiesel production via transesterification of soybean oil.

The experimental results confirmed that catalyst composition plays a critical role in determining biodiesel yield. The 70:30 activated carbon to Ca-Fe oxide ratio exhibited the best performance, achieving a maximum biodiesel yield of 92%, indicating an optimal balance between active catalytic sites and support material.

The study also showed that while catalyst composition significantly influenced biodiesel conversion efficiency, its effect on fuel energy content (HHV) was minimal, with all samples showing relatively similar values within the biodiesel range. Fuel property analysis further indicated that the produced biodiesel generally met acceptable cold-flow characteristics, although density values were slightly above standard EN 14214 limits.

From a broader perspective, the integration of CaO, Fe_3O_4 , and activated carbon provided a multifunctional catalyst system combining catalytic activity, structural stability, and improved surface characteristics. The presence of Fe_3O_4 also introduced potential advantages for catalyst recovery due to its magnetic properties.

Overall, the research objectives were successfully achieved, and the findings confirm that Ca-Fe oxide supported on activated carbon is a promising heterogeneous catalyst for biodiesel production with potential for further optimization and scale-up.

6.4 Recommendations

Based on the experimental findings of this study, several recommendations are proposed to improve catalyst performance and overall biodiesel production efficiency.

Firstly, further optimization of the catalyst composition is recommended. Although the 70:30 activated carbon-oxide ratio showed the highest biodiesel yield in this study, intermediate ratios and a wider compositional range should be investigated to identify a more precise optimum.

Secondly, additional operating parameters such as reaction time, reaction temperature, methanol-to-oil ratio, and catalyst loading should be systematically optimized. This would improve process efficiency and provide a more robust understanding of reaction behavior.

Thirdly, catalyst reusability and long-term stability should be evaluated through multiple reaction cycles. This is essential to assess the practical applicability of the catalyst in industrial biodiesel production.

Furthermore, improving catalyst synthesis conditions such as calcination temperature, reactor, impregnation time, and precursor concentration may enhance active site dispersion and improve catalytic efficiency.

Finally, purification steps for biodiesel should be improved to ensure compliance with international standards, particularly with respect to density and impurity removal.

6.5 Scope for Future Work

The present study on Ca-Fe oxide supported on activated carbon for biodiesel production provides a strong foundation for further research and development in heterogeneous catalysis for renewable fuel production. Future work should focus on evaluating catalyst reusability over multiple reaction cycles to better understand long-term stability and deactivation behavior. This will be essential for assessing industrial applicability. Pilot-scale or continuous reactor studies are also recommended to bridge the gap between laboratory-scale experiments and industrial implementation. This would provide more realistic insights into process scalability and economic feasibility.

Further research may also explore the use of alternative feedstocks such as waste cooking oil, non-edible oils, or algal oils to improve sustainability and reduce feedstock cost. Advanced characterization techniques such as X-ray Photoelectron Spectroscopy (XPS), Transmission Electron Microscopy (TEM), and Temperature-Programmed Desorption (TPD) are recommended to gain deeper insight into catalyst surface chemistry and active site behavior.

Finally, development of environmentally sustainable catalyst synthesis routes, including the use of waste-derived activated carbon or low-energy preparation methods, can further enhance the green chemistry aspect of biodiesel production.

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Appendix

Appendix A : Mapping of the research problem with the attributes of complex engineering problems

Table 6: Knowledge profile of the research

Attribute	Applicable/ Not Applicable	Remarks
K1	Yes	Understanding transesterification process, co-precipitation, metal oxide formation.
K2	No	-
K3	Yes	Catalyst synthesis, wet impregnation technique, drying and calcination processes.
K4	Yes	Depth in chemical reaction engineering, catalysis, and materials science.
K5	No	-
K6	No	-
K7	Yes	Understanding biodiesel as a renewable energy source and its importance in future fuel alternatives.
K8	Yes	Engages with current research on nanostructured Ca-Fe oxide catalysts and their application in biodiesel production.

Table 7: Complex engineering problem solving

Attribute	Applicable/ Not Applicable	Remarks
P1	Yes	The problem involves catalyst preparation which requires in-depth knowledge of chemical engineering principles (K3), specialized understanding of catalysis and materials science (K4), and application of recent research findings in catalyst development (K8).
P2	Yes	The research must balance surface area, metal loading, thermal stability, and catalytic activity, which can conflict during synthesis and calcination.
P3	Yes	The process requires abstract thinking and experimentation to optimize catalyst structure and performance.
P4	No	-
P5	No	-
P6	No	-
P7	Yes	The problem includes multiple interrelated components: nanoparticle synthesis, catalyst preparation and yield analysis.

Appendix B : Research plan (preliminary and revised)Preliminary Work Plan

Preliminary Work Plan

ID	Activities	Work Plan													
		Level 4/Term 1 (14 Weeks)													
		1	2	3	4	5	6	7	8	9	10	11	12	13	14
1	Literature Review														
2	Raw Material and Sourcing														
3	Synthesis of Fe ₃ O ₄ Nanoparticles														
4	Fe ₃ O ₄ /AC composite preparation														
5	CaO impregnation on Fe ₃ O ₄ /AC														
6	Calcination and Catalyst Activation														
		Level 4/Term 2 (Tentative 14 weeks)													
		1	2	3	4	5	6	7	8	9	10	11	12	13	14
7	Catalyst Characterization														
8	Biodiesel Reaction Setup														
9	Reusability and Magnetic Recovery Tests														
10	Final and Yield Analysis and Comparisons														
11	Data Analysis and Interpretation														
12	Drafting Thesis Chapters														
13	Review, proof reading and final editing														
14	Final submission and Defense Prep														

Revised Work Plan

ID	Activities	Work Plan													
		Level 4/Term 1 (14 Weeks)													
		1	2	3	4	5	6	7	8	9	10	11	12	13	14
1	Literature Review														
2	Raw Material and Sourcing														
3	Synthesis of Fe ₃ O ₄ Nanoparticles														
4	Fe ₃ O ₄ /AC composite preparation														
5	CaO impregnation on Fe ₃ O ₄ /AC														
6	Calcination and Catalyst Activation														
		Level 4/Term 2 (Tentative 14 weeks)													
		1	2	3	4	5	6	7	8	9	10	11	12	13	14
7	Catalyst Characterization														
8	Biodiesel Reaction Setup														
9	Reusability and Magnetic Recovery Tests														
10	Final and Yield Analysis and Comparisons														
11	Data Analysis and Interpretation														
12	Drafting Thesis Chapters														
13	Review, proof reading and final editing														
14	Final submission and Defense Prep														

Appendix C : Risk Assessment

Assessment details

Location: Old Academic Building (OAB), Bangladesh University of Engineering and Technology (BUET)

Room No.: Kinetics Engineering Laboratory

Department: Chemical Engineering

Faculty: Chemical and Material Sciences

Other Information:

Assessment Date: 5 July, 2025

Task/Process Description: Ca-Fe Oxide on Carbon Catalyst for Biodiesel Production

Assessment Team:

Name and Designation	Role	Signature
Md Al Amin Sajid	Preparer	
Ahasan Habib Rafy	Preparer	

Table 8: Table For the risk assessment of the project

Hazard No	Hazard Description	Consequences & Likelihood	Risk	Corrective Actions/ Risk Controls	Timing & Responsibility
1	Handling of hydrochloric acid (HCl) for activated carbon washing - corrosive, causes burns, inhalation risk	Moderate & Likely	High	Engineering: Use fume hood Administrative: SOP training and supervision PPE: Acid-resistant gloves, goggles, lab coat, enclosed shoes	During acid washing; experimenter
2	Handling of ammonium hydroxide (NH ₄ OH) during Fe ₃ O ₄ synthesis - corrosive, ammonia vapor release	Moderate & Possible	Medium	Engineering: Work inside a fume hood Administrative: Use diluted solutions; SOP adherence PPE: Gloves, goggles, lab coat	During Fe ₃ O ₄ synthesis; experimenter
3	Handling of ferric nitrate [Fe(NO ₃) ₃] and ferrous nitrate [Fe(NO ₃) ₂] - oxidizers; cause eye, skin, and respiratory irritation; may intensify fire	Moderate & Possible	Medium	Engineering: Store away from combustibles and flammables; use ventilated workspace Administrative: SOPs for oxidizer handling PPE: Gloves, goggles, lab coat,	During Fe ₃ O ₄ nanoparticle synthesis; experimenter

				dust mask if powder form is used	
4	Heat release and splashing during mixing of $\text{Fe}^{2+}/\text{Fe}^{3+}$ with NH_4OH - risk of local overheating and spillage	Moderate & Unlikely	Medium	Engineering: Add NH_4OH slowly under stirring Administrative: Monitor temperature; avoid closed vessels PPE: Face shield, gloves, apron	During Fe_3O_4 nanoparticle synthesis; experimenter
5	Handling of calcium nitrate $[\text{Ca}(\text{NO}_3)_2 \cdot 4\text{H}_2\text{O}]$ - oxidizer, may intensify fire; causes skin and eye irritation	Moderate & Possible	Medium	Engineering: Keep away from flammables; work in ventilated area Administrative: SOP for weighing and handling oxidizers PPE: Gloves, goggles, lab coat	During wet impregnation and drying; experimenter
6	Use of drying oven and furnace (100-600 °C) - burn or fire hazard	Moderate & Possible	Medium	Engineering: Use heat-resistant gloves Administrative: SOP and labeled zones for hot surfaces PPE: Heat-resistant gloves,	During drying and calcination; experimenter

				lab coat, closed shoes	
7	Disposal of chemical waste (HCl, NH ₄ OH, metal solutions) - environmental hazard if improperly handled	Moderate & Possible	Medium	Administrative: Use labeled chemical waste containers, follow institutional disposal procedures.	After experiments; experimenter.
8	Methanol use in transesterification - highly flammable, toxic if inhaled or ingested	Major & Possible	High	Engineering: Ensure adequate ventilation Administrative: Keep away from ignition sources; store in flammables cabinet PPE: gloves, splash goggles, flame-resistant lab coat	During transesterification ; experimenter

Appendix D : Environmental Mitigation and Monitoring Plan

Table 9: Table for the environmental mitigation and monitoring plan.

Activity description	Identified environmental aspects or impacts	Mitigation measure(s)	Monitoring indicator(s)	Monitoring and reporting frequency	Responsibility	Field monitoring / issues
Preparation of Fe ₃ O ₄ nanoparticles	Generation of chemical wastewater; ammonia fumes.	Conduct reaction under fume hood; neutralize wastewater before disposal	pH of waste solution; presence of fumes.	During each experiment.	Experimenter	pH of liquid waste was successfully adjusted to 7.0 before disposal.
Catalyst synthesis (Ca-Fe/AC impregnation)	Chemical handling risk; solvent waste.	Use PPE; collect solvent waste separately.	Proper labeling of waste containers.	During each experiment.	Experimenter	Waste managed as per lab protocol.
Calcination of Catalyst	Energy consumption and potential emission of fumes.	Use energy-efficient furnaces.	Furnace run-time logs.	Weekly equipment check.	Experimenter	Furnace used as per lab protocol.
Biodiesel reaction testing	Methanol vapor emission; liquid waste-glycerol.	Recovery of methanol via condensation ; proper disposal of glycerol byproduct.	Odor/vapor presence.	Per reaction cycle.	Experimenter	Controlled conditions maintained.
General laboratory activities	Solid waste (filter paper, gloves, mask).	Segregated waste disposal..	Waste bin inspection	Weekly	Lab authority/ Experimenter	Acted as per lab protocol.